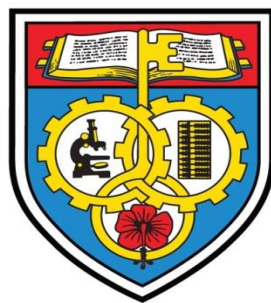


Facility Booking System

By

Ong Zhe Zhun



TARU
TUNKU ABDUL RAHMAN
UNIVERSITY COLLEGE

FACULTY OF COMPUTING AND
INFORMATION TECHNOLOGY

TUNKU ABDUL RAHMAN UNIVERSITY COLLEGE
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Facility Booking System

By

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Supervisor: Ms. Choy Lai Fun

A project report submitted to the
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in partial fulfillment of the requirement for the
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Department of Information and Communication Technology

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Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University College rules and regulations and would be subjected to disciplinary actions.

Ong Zhe Zhun

Bachelor of Information Technology (Honours) in Software Systems Development

Abstract

In such a modern era where technology is getting more and more advanced, there is a trend where people start to look for and more emphasize on an efficient and effective method that able to bring a lot of convenience to people. The efficient and effective method that could bring more convenience to people includes software that could be found such as Joan's Meeting Board, a Facility Booking System (FBS), and so much more. Following the introduction of the Internet in November 1995, Internet users have increased significantly to date, occupying 76.9 percent of Malaysia's population, making Malaysia an Internet-oriented society (Twenty-one, 2017), most of which have smartphone access to the Internet, which is the current trend in Malaysia (Osman *et al.*, 2012). Unfortunately, it is hard to say that everyone is able to adapt to the rapidly changes of technology trend and applying those advance technology to everywhere. There are still some places which unable to catch up to follow the trend and retain the traditional method to perform tasks. But luckily Tunku Abdul Rahman University College (TARUC) is on the way catching up with the technology trend. College students are putting their effort creating a facility booking system to make the facility booking process and experience more convenient. The existing booking system done allows admin user to make facility booking, customer and staff account management, event management, and make announcement. However, there are some vulnerabilities left behind by students that done this project and also still much room for further enhancement and towards the existing facility booking system done. The proposed enhancement allows admin user to plan the schedule of facility and resources or venue availability before allows customer to make booking, control the accessibility of admin user with predefined role, maintaining customers' record, additional api support for mobile development and vulnerabilities fixing for smooth facility booking, announcement management, and service and rules management experience.

Acknowledgement

Firstly, I would like to express my special thanks of gratitude to my supervisor Ms. Choy Lai Fun. She had liable to supervise my progress in every chapter of documentation. Besides that, she provided me with lots of useful feedback and correct me if I do anything wrong so that I can make necessary amendment to my project. She had always reminds me with the submission deadline of each project chapter, so that I can submit every chapter on time.

Next, I would like to express my special thanks of gratitude to Mr. See Kwee Tek, who gave me the golden opportunity to take over the challenging yet wonderful Smart campus final year project which is Facility Booking System from seniors.

Besides that, I would like to express my special thanks of gratitude to my team partner Melvin Low Eik Kit. He had always assist me when I faced problem and encourage me a lot when I faced problem emotionally

Last but not least, I would like to express my special thanks of gratitude to my parents. They provided me with financially and emotionally support. Without my parents, I would not be able to focus myself and complete my final year project.

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Chapter 1

Introduction

Chapter 1 Introduction

In such a modern era where technology is getting more and more advanced, there is a trend where people start to look for and more emphasize on an efficient and effective method that is able to bring a lot of convenience to people. After the internet was introduced, the impact is starting to affect everything around the globe and started to make computerized systems where technology is getting more advanced and give a lot of convenience to people and also a booster to the world business market (Molchanova, 2014). The efficient and effective method that could bring more convenience to people included those software that could be found everywhere such as Purchase Ordering System (POS), Transaction Processing System (TPS), and so much more.

Sadly, it is difficult to say that everyone can adapt to the rapidly evolving technological trend and apply this advanced technology everywhere. There are still places that cannot keep up with the trend and keep the traditional method of performing tasks. The most obvious ones will be the facility booking method in education institutions. There are people sharing thoughts where they found that the facility booking and confirmation process is very inconvenient and time consuming. The traditional facility booking process is that, for example, students or staffs (customers) in TARUC want to make a booking of badminton court for their co-curriculum activity, they need to walk far distance to the booking counter at sport complex to make the booking possible. The entire process is inconvenient, slow and inefficient due to requestors are required to physically approach the booking counter which was located far away from where people usually went to. Besides, loss of data might happen because all the booking information was written black and white. It will be frustrated when our data storage is visited by unexpected hazards such as fire disasters. The amount of booking paper stored can be very messy and it takes time to do arrangement. During the clearing of old data, mainly at the end of the year, data that exceed 5 years no longer required to be checked are deleted. Recent data might mix with the old ones and dispose. Thus lead to serious issue which is loss of data.

In order to cater the situation, students, mainly final year students that working for their final year project or internship employee in TARUC has been trying to develop a facility booking system to solve the problem. But there are still some minor issue, problem and vulnerabilities to be solved and additional features to be added before the launching of the

system. Therefore, this project is to do some further enhancement and some tweak towards the Online Facility Booking System. The enhancement will focus on the back-end, which is the web online booking. The enhancement include functions such as admin role management, venue and customer management and additional api support for new platform development. Vulnerabilities fixing has been carried out towards facility booking, announcement management and service and rules management modules.

The most beneficial target for using the proposed enhancement will be staffs that was assigned to manage the facilities in tertiary institutions. Staffs able to manage the facility schedule to limit the facility availability before open for booking, having better control towards staff accessibility of modules, able to have better control towards customer information, and better and smoother user experience in facility booking, announcement management, and service and rules management procedure.

1.1 Objective

The main objectives of this project is to increase the working efficiency of facility management staffs. The available booking system requires a staff stayed, sitting at the booking counter to process booking from customer (students and staffs). In addition, the speed of staff process the booking is relatively slow, and hence results in a queue of customer.

Besides, this is also meant to provide a better platform for staffs to perform and process facility booking. More and more people and companies are transitioning their workflow from manually become digitalize, because of the processing speed of a computer is way faster compare to human (Fletcher, Anthony, 1986). They prefer to use mobile phone to perform as much task as possible since they are using smartphone as a daily driver. This helps them to eliminate the need to physically attend to the location and become daily life needs (Osman *et al.*, 2012).

1.2 Background

Tunku Abdul Rahman University College (TARUC) is a premier tertiary educational institution with more than 49 years of excellence in providing quality and affordable education. Known then as Tunku Abdul Rahman College (TARC) since its inception in

1969. The institution has grown from strength to strength offering pre-university, diploma, bachelor degree as well as postgraduate programs.

Currently, Tunku Abdul Rahman University College (TARUC) is using traditional booking method, which is using black and white paper to process and record students and staffs' (customer) booking information. There are many limitations to be found with the use of traditional booking method. Firstly, the booking process is slow and inefficient. Since the booking schedule of every facility is recorded on paper separately, staff need to search through huge amount of paperwork to find the specific facility booking record.

There are several competitors found to be with the facility booking system. The first will be the Meeting Board developed by Joan. The system is mainly focusing on managing meeting room's schedule. The system able to sync the meeting room schedule with existing calendar from application such as Google Calendar via their online portal. Then they also implemented dedicated hardware to place in front of the meeting room to show the meeting room booking schedule.

The other competitor found will be Skedda, a venue booking and scheduling system. Skedda uses a booking calendar to manage the booking schedule of facility. The booking calendar even able to sync up with other calendar app such as Google Calendar and others. The user interface of the system is well-designed, even able to sync up with their mobile application to perform booking anytime, anywhere.

1.3 Advantage and Contribution

In traditional facility booking process, students and staffs are required to physically approach the booking counter to make any facility booking with related staffs at the counter. With the proposed system and enhancement applied, the time spent on facility booking are expected to reduce in a significant figure. Certainly there are always competitors in the society, company such as Planyo are focusing on automated system for their booking system that require zero interaction with human-beings and yet the physical access of the system do not require any internet connection to access certain place customer booked remotely. Besides, there are also company that was succeed in the facility booking areas such as Joan Meeting Boards. The setup process for the boards is simple, yet the scheduling is directly linked to Google Calendar. Any booking made on Google Calendar will be updated immediately.

1.4 Project Plan

1.4.1 Functional Requirement

- a) Admin Module
 - Restrict admin to access module based on his/her role assigned.
 - Allow admin to reset their password.
- b) Announcement Module
 - Vulnerability fixing.
- c) Booking Module
 - Vulnerability fixing.
 - Algorithm changing for additional filtering
- d) Customer Module
 - Allow admin to help customer perform account registration.
 - User profile to provide more user information.
 - To allow admin to import a list of customer record.
 - Allow customer or admin to perform password reset.
- e) Rules and Service Module
 - Vulnerability fixing.
- f) Venue Module.
 - Used to store facility and resources information before allowing the scheduling of facility.
 - Allow staff to manage the operating hour of the venue

1.4.2 Non-Functional Requirement

- a) Availability
 - The system must ensure that it can operate 24/7 without interruption of server problems or bugs that affect the system's availability.
- b) Security and privacy
 - Security and privacy will be the most important requirement in the system as it involved lots of important information such as student and staff's IC number, name, phone number etc.
- c) Reliability

- The system must make sure every task performed is right and accurate to prevent inaccurate information for misleading
- d) Performance
- The system must be powerful enough to cater different kind of processes since it involves a lot of algorithms, and also able to handle lots of request at the same time. An amount of more than 20,000 user at peak hour are expected to access the system at the same time.
- e) Usability
- The system are expected to be efficient, effective and also provide satisfaction which the user able to use the system properly and increase user's satisfaction at the same time.

1.4.3 System Requirement

- a) Hardware Components
- Desktop computer with Windows 7 and above
 - Internet
 - Smart Phone
 - Server
 - QR Scanner
- b) Software Requirements
- Netbeans IDE
 - Visual Studio Code
 - Android Studio
 - SQL Database
 - Node.js
 - React Native
 - C# Programming Language
 - Java Programming Language

1.4.4 Milestone

Table 1.1 Project Milestone

Phase	Milestone	Deadline
System Planning	First draft proposal submission	05/06/2018
	Final proposal submission	22/06/2018
System Analysis System Design	Introduction	13/07/2018
	Research background	10/08/2018
	Methodology & Requirement analysis	15/10/2018
	Front-End System App UI Design (Students)	17/10/2018
	Front-End System Database design	25/10/2018
	Front-End System Module design (Students)	31/10/2018
	Submission of Project I Portfolio	16/11/2018
System Development	Front-End System App Coding (Students)	18/11/2018
	Unit Testing (Students)	15/01/2019
	Module Testing (Students)	17/01/2019
	Integration Testing	09/02/2019
	Final System Testing	04/03/2019
	Submission of Draft FYP Report	29/03/2019
	Submission of Final FYP Report	05/04/2019

1.4.5 Software Development Plan

Process software models consist of different processes or methodologies selected according to the goals and objectives of the project. The question of selecting software process models should be considered very carefully. After several discussion made, prototype model was selected as the software process model of the project. Prototyping is a popular software process model whereby it able to get or further clarify the valuable customer requirement at the early stage of development. A clear customer feedback helps developer to understand what customer expected from the product.

According to figure 1.1, figure 1.2 and figure 1.3, a total of four main phases or activities are involved in the prototyping model. The first phase is requirement gathering, which is to gather all the requirement as detail as possible. Requirement gathering might able to be done by several ways such as interview, questionnaire, research and others. The second phase is quick design, which will based on the requirement gathered and design the preliminary layout of the software that provide the approximation of output required. Then an initial prototype will be developed based on the preliminary layout. Next phase will be prototype evaluation by client reviewing the preliminary layout from the quick design phase. After client check the output, client will give some opinions towards the prototype. Developers will collect those opinions and examine the part that need to improve or discard. Last but not least will be refining the end functionality by modifying the preliminary layout according to feedback from client evaluation and develop a new version of prototype. The software will be modified in a routine based until the prototype is satisfied by client.

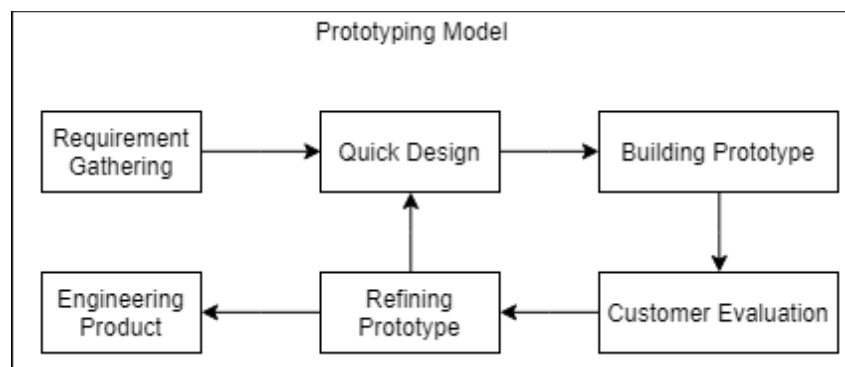


Figure 1.1: Steps of Prototyping Model

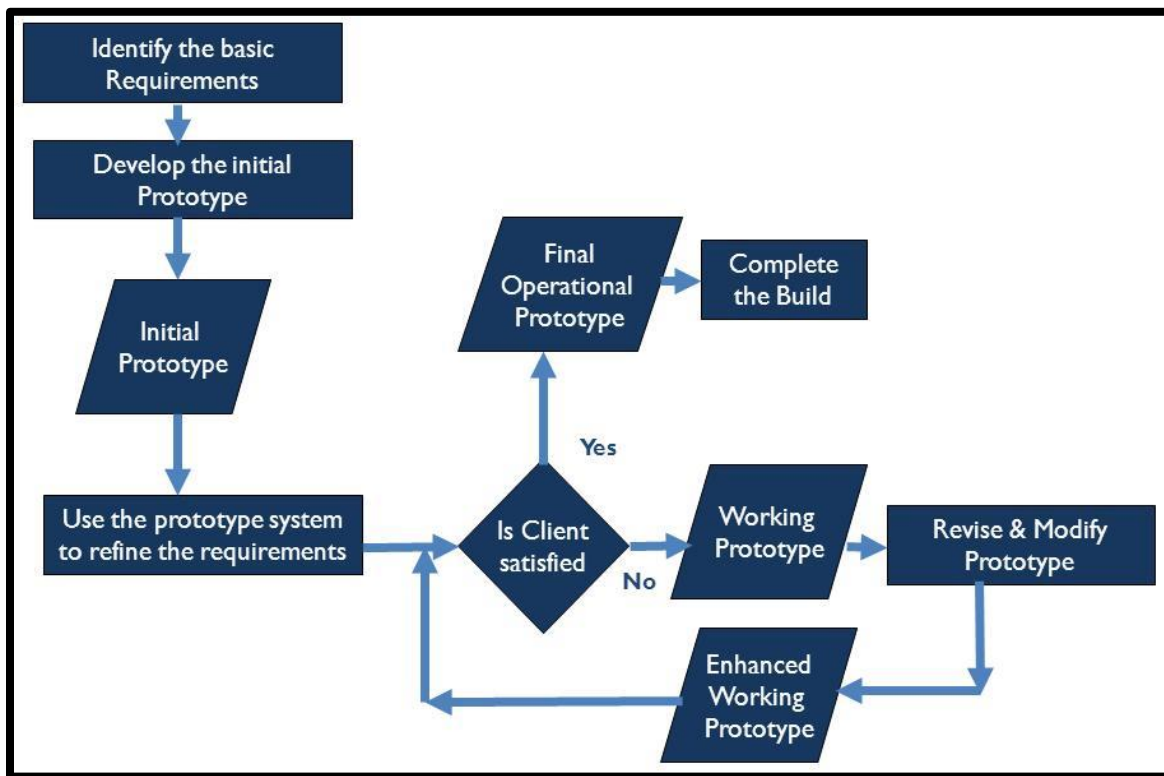


Figure 1.2: Detailed Step of Prototyping Model

There are numerous reasons that prototyping model be the most suitable choice for the project. Firstly, user are actively involved in the development. User involvement is needed throughout the whole prototyping development period. This allows user to see the partial product at the early stage of development. Developer able to interact with the prototype and obtain immediate feedback from client and end-user. This ensures users to have the greatest level of satisfaction and comfort when using the system. The most important is that prototype able to eliminate misunderstanding between developer and end-user during the development process.

Besides, it reduces time and cost as the defects can detect much earlier. Developers able to detect defects and take action to improve it much earlier, therefore the software developed is less expensive and the development period is shorter. The quality of the software is also able to be enhanced. If any defects are found after the software implementation, the cost and time incurred to maintenance will be exponentially more.

Furthermore, prototype model allows better quality system to be delivered. New requirements and potential defects can be easily identified at the refinement stage, which lead to building a better quality software. The clear user requirements collected eliminate ambiguities and miscommunication, so the software is developed with better quality. The risk of defect associated with the software is reduced.

Last but not least, missing or confusing functionality can be identified easily. The developing prototype will be compared to software specification from time to time. If missing or confusing functionality is detected. The prototype system allows user to provide the feedback and review to developers because prototyping model provide the actual look and feel of the software. This help developer identifies the problems.

1.5 Team and Organization

The Online Facility Booking System consists of front end and back end module. Each module will be handle by one member. Melvin Low Eik Kit is responsible for front end and Ong Zhe Zhun is responsible for back end.

Table 1.2: Table of workload distribution

Module and Sub-modules	Melvin Low Eik Kit	Ong Zhe Zhun
Front End	✓	
Authentication	✓	
Announcement	✓	
Schedule booking	✓	
Booking Record	✓	
Friend system	✓	
Back End		✓
Admin		✓
Announcement		✓
Booking		✓
Customer		✓
Rule and Service		✓
Venue		✓

1.6 Thesis Outline

Chapter 1 Introduction

- 1.1 Objective
- 1.2 Background
- 1.3 Advantages and Contribution
- 1.4 Project Plan
- 1.5 Project Team and Organization
- 1.6 Thesis Outline
- 1.7 References

Chapter 2 Literature Review

- 2.1 Project Background
- 2.2 Literature Review
- 2.3 Feasibility Study
- 2.4 Chapter Summary and Evaluation
- 2.5 References

Chapter 3 Methodology and Requirement Analysis

- 3.1 Methodology
- 3.2 Requirement Analysis
- 3.3 Chapter Summary and Evaluation
- 3.4 References

Chapter 4 System Design

- 1.1 System Design
- 1.2 Chapter Summary and Evaluation
- 1.3 References

Chapter 5 Implementation and Testing

Chapter 6 Discussions and Conclusion

- 6.1 Summary
- 6.2 Achievement
- 6.3 Contributions
- 6.4 Limitations and Future Improvement
- 6.5 Issues and Solutions

1.7 Chapter Summary and Evaluation

When I was doing this chapter 1, I able to gain more overview towards the whole project, but there are still missing of information here and there that need to further elaborate the details and provide more insight about this newly proposed system with enhancement. The part that I most struggle with is to identify the most suitable enhancement for the system or identifying the vulnerabilities or bug leftover from existing completed modules. This chapter explain about how the project team organizing their workload, the background of the related client and competitors against proposed system, and also the advantage and contribution of proposed system to society.

Chapter 2

Literature Review

Chapter 2 Research Background and Related Works

2.1 Company Background

Tunku Abdul Rahman University College (TARUC) is a leading tertiary school with over 49 years of excellence in quality and affordable education. Formally known as Tunku Abdul Rahman College (TARC). The institution has grown strongly by offering pre - university, diploma, bachelor's degree and postgraduate programs.

To date, TARUC has over 27,000 students studying at its main campus in Kuala Lumpur and branch campuses in Penang, Perak, Johor and faculty branches in Pahang and Sabah. Since 1969, TARUC has produced more than 180, 000 graduates.

2.2 Project Background

Online Facility Booking System

The online booking system allows people to reserve a certain venue at a certain time (e.g. badminton court 1) to perform activities (e.g. play badminton), online booking can be done by browsing the website or by using the mobile app developed by developers. The app or web can allow people to book with just one click or touch that saves a lot of time and effort compared to the manual booking system. There are currently 7 modules on the front end, while 5 modules are on the back. The first back-end module will be the scheduling, the module is created to pre-allocate certain faculties and resources for important events such as holidays, and etc. This to prevent customer from booking during the period. Next will be the customer module, the module is focusing on registration confirmation after customer register their account via mobile apps. It also extend to store more customer personal information so that able to be used during emergency. The third is admin role management module, the module will monitor each user's role assigned and redirect to module that the user was allowed to access. The forth module is venue module, the module stores all the facility and resources information that open for booking. All the facility and resources must be entered into the system before proceed for booking. The last module will be the notification module, the module will detect any changes, cancellation or announcement and send the information to customer to notify them via notification or email.

2.2.1 Literature Review (Existing Similar System)

1. Planyo



Figure 2.1 shown Planyo company logo

Planyo is an online booking software hosted on their dedicated servers with daily backup and real - time database replication using a modern web interface with features of more than 30 languages, including features that can send email and text messages to their customers and provides more than 50 payment gateways such as PayPal, Visa, MasterCard and so on. The pricing of the system is flexible starting from RM90.00, It will be charged based on the number of features requested and the specific system needed, such as the booking system for the bus shuttle and the online appointment system.

2. Joan Meeting Boards

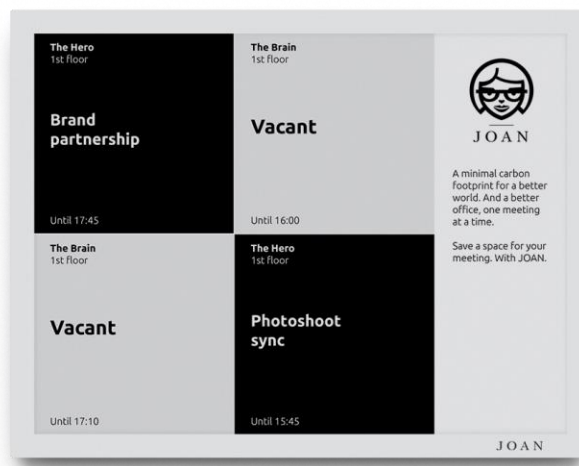


Figure 2.2 shown Joan Meeting Board

Joan is an online booking system, solution for every office space. It is an easy-to-use device that helps companies organize and reserve meeting spaces and huddle rooms better. Joan comes in a combination of meeting boards that placed outside of meeting room to show the meeting room booking schedule, an online portal to control all the schedule, and also existing calendar to sync the schedule over a secured internet or Wi-Fi connection.

2.2.2 Potential Market and Target Users

The application is mainly targeting users which is in tertiary educational institution. User such as students, staffs and alumni will be the most beneficial target from the proposed system because the necessity of approaching respective facility booking counter was eliminated. It was quite frustrated when approaching counters and found that the booking slot available left do not fits own schedule. Furthermore, user feels inconvenient to go to booking counter make their booking. Additionally, student need to walk far distance to the booking counter. User to access the booking system by using the mobile application (front-end) to make their facility booking. The booking can be done anywhere within the predefined available (advance booking) time-frame. User also able to approach booking counter to make their booking where admin will help user to perform booking process via web application (back-end).

2.3 Feasibility Study

1. Organizational Feasibility

TARUC is recommended to deploy the server via cloud services, the main reason being that it requires high installation and maintenance costs. The price of a powerful server hardware that is available to more than 20,000 users at the same time is on average of more than RM2500.00. Besides, power consumption to support the server operation is very high. The server had to be located in a cool room that was operational around the clock, and it needed a lot of power to power up the server and its load. In addition, the cloud server, which is part of fog computing, is seen as the intermediary for connecting to the Internet of Things (IoT) by classifying IoT applications and planning computing resources (Chen *et al.*, 2017). Furthermore, training needs to be provided to train employees to maintain the server system, which could cost a fortune. Cloud server help by leveraging existing infrastructure such as PC, laptop, smartphone and tablet as a terminal to access the system. TARUC management

is then able to decide whether it is necessary to subscribe to the service without worrying about the problems mentioned above.

2. Operational Feasibility

The proposed system will provide staff with a simple but modern and user-friendly interface. The user first logs on to his or her own account because the information is synchronized with the intranet system. The user can make booking based on customer requirements or process the booking request made by the customer. There will be a schedule list allowing users to select, system will prompt error if the schedule is not available. A QR code is generated to the customer on the spot or sent to the customer's account for check-in purposes.

2.4 Chapter Summary and Evaluation

In this chapter, the information for the project goes deeper and deeper, even from a different perspective the system needs to be discovered and elaborated. Furthermore, there are too many competitors to be found in the market

During the process of work, it become more and more difficult to find a good journal that related to the topic. The information to be given for the project is going deeper and deeper, even need to discover the system from different perspective and further elaborate it. In a nutshell, this chapter provides information such as research conducted towards the competitors of the system, its feasibility study, and etc.

Chapter 3

Methodology and Requirement Analysis

Chapter 3 Methodology and Requirement Analysis

3.1 Methodology

The prototype methodology is the model for software development in my project. This methodology is much more practical and suitable for the time we are in the 21st century because of the rapid technological trend that could lead to changes in demands. The advantage of the prototype model is that the cost is low, because the requirement is not really well defined at first, but encourages the involvement of customers and developers, which means that my team can discuss more requirements if there is a missing or unsatisfied part after the prototype has been created and then show the customer who is the Department of Student Affair (DSA). The prototype model also emphasized fast and low costs, since the prototype model only focuses on the requirements of the student department, then refines the prototype further and finally becomes a final product that can be supplied.

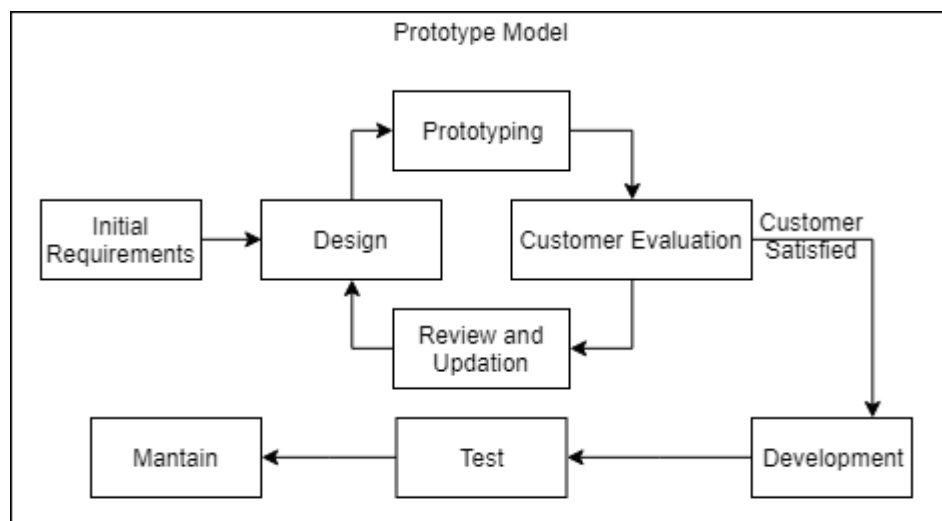


Figure 3.1 shown the Phases of Prototype Software Process Model

In the beginning, our team performed an initial collection of requirements from the department of student affairs, then we developed the user interface (UI) and created some relevant diagrams to facilitate the work during the coding phase during the prototyping process. Once the prototype has been completed, we will demonstrate it to the DSA and collect some reviews and updates necessary for the prototype changes. After DSA is satisfied with our prototype, we start the development phase for setting up a server, reorganizing the code, etc. Then, test the product in a controlled environment before it is

released to the public. When the test is completed, the product can be delivered and the system must be kept from time to time.

3.2 Requirement Analysis

A) Literature Review

The source can be found during the collection of literature research information, but most sources are unreliable because they are not a journal or an article approved by researchers. Using a source without citation could be inaccurate and affect the quality of the project. Citation is proof of your work. It is not based on assumptions, but on the reference of certain materials when you do your work (MacRoberts and MacRoberts, 2010). To solve the problem, I need to find the source of Google Scholar and also ScienceDirect, which is recommended by TAR UC Library and contains lots of magazines, books and etc. The Harvard Referencing System was properly used to cite the journal during the literature review. To do this correctly, Mendeley Desktop was used to facilitate the referencing of literature, it is easier to use and is also suitable for desktop and laptop. It is also available to install the Microsoft Word plugin in Mendeley so that you can insert citations while writing the literature review.

B) Observation

Observation is a method by which the analyst visits the environment himself, observes and understands the environmental workflow. For reference purposes, existing similar systems are downloaded, tested and analyzed. This is to obtain information such as the validation and operation of existing system and can be included in the project. The list of observed systems is listed below.

- i) Planyo (Valid for 30 days free trial, from <https://www.planyo.com>)
- ii) Joan Meeting Boards (Contacted friends that do have one to experience it)

3.3 Fact Recording

Facts recording is one of the important processes for seeing the system in a very large picture or in very detail. A data model can be assumed as a flow chart that shows the relationship between different data. The process is time consuming, but it must be because it is a very important step. The methods used for fact recording are described as below.

A) Use Case Diagram

The use case diagram used in project shows how specific actors take the actions (use case). This ensures that the system's functional requirement is met. The use case diagram shows clearly the interaction between the user and the function of the modules in the system. Use case description to explain the flow of the use case.

B) Activity Diagram

The process of the system can be illustrated with an activity diagram. The diagram can represent a whole system's activity flow and capture the system's behavior. It can also describe the sequence and the parallel, connected and simultaneous system flow from one activity to another. Information such as the user involved, system processes, data required in the process, and the interaction or flow of workflow between user, system and data stored was identified.

3.4 Fact Analysis

The technique used for data analysis is the qualitative data analysis technique. Qualitative data analysis is the process of extracting data from the explanation and interpretation of journals, articles, books or even interviews to reduce data in order to find the best solutions to the research question or the collection of requirements. Throughout the analysis process, some effective and simplified solutions are found to reduce the complexity of system, such as a more secure database communication method, password recovery method, Simple Network Management Protocol (SNMP) usage and etc.

3.5 Requirement Analysis

Project Scope

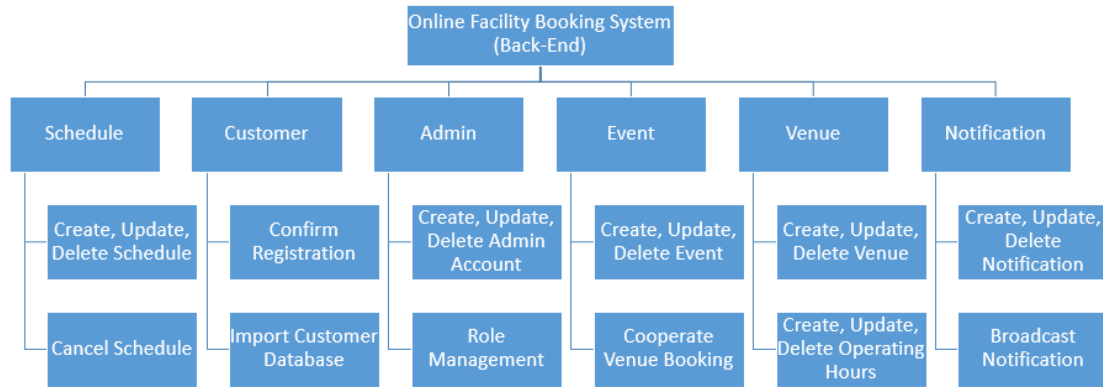


Figure 3.2: Project Scope for Online Facility Booking System

3.5.1 Development Environment

Table 3.1: Hardware Specification for development purpose' computer

Laptop Model	Dell Inspiron 14 3000 Series
Processor	Intel(R) Core(TM) i7-5500U CPU @ 2.40GHz
Operating System	Microsoft Windows 10
Display	15.6" Display
Memory	8GB
Storage	1TB

Table 3.2: Software and Programming Language Used

Software	Browser – Google Chrome Operating System - Microsoft Windows Documentation - Microsoft Word 2013, Adobe Reader PDF, Gmail, IBM Rational Software Architect for WebSphere Software Development – React Native 0.57, Visual Studio Code, Android Studio 3.2.1
Programming Language	Java, HTML

3.5.2 Operation Environment

Table 3.3: Recommendation for Smartphone

OS version	Android OS v8.0 and above
Processor	2.0 GHz
Memory	2GB RAM
Services	Wi-Fi Capability, GPS

3.5.3 Functional Requirement

The functional requirement of Online Facility Booking System (Back-end) are as follows:

- a) Admin Module
 - Restrict admin to access module based on his/her role assigned.
 - Allow admin to reset their password.
- b) Announcement Module
 - Vulnerability fixing.
- c) Booking Module
 - Vulnerability fixing.
 - Algorithm changing for additional filtering
- d) Customer Module
 - Allow admin to help customer perform account registration.
 - User profile to provide more user information.
 - To allow admin to import a list of customer record.
 - Allow customer or admin to perform password reset.
- e) Rules and Service Module
 - Vulnerability fixing.
- f) Venue Module.
 - Used to store facility and resources information before allowing the scheduling of facility.
 - Allow staff to manage the operating hour of the venue

3.5.4 Non-Functional Requirement

The non-functional requirements of Online Facility Booking are as follows:

- a) Availability
 - The system must ensure that it can operate 24/7 without interruption of server problems or bugs that affect the system's availability.
- b) Security and privacy
 - Security and privacy are the most important requirements in the system, as they involve a lot of important information, such as IC number, name, telephone number, etc. for students and staff
- c) Reliability
 - The system must ensure that every task performed is correct and accurate to prevent misleading inaccurate information.
- d) Performance
 - The system must be powerful enough to accommodate different types of processes because it involves a lot of algorithms and can handle a lot of requests simultaneously. At peak hour, over 20,000 users are expected to simultaneously access the system.
- e) Usability
 - The system is expected to be efficient, efficient and also satisfied that the user can use the system properly and simultaneously increase user satisfaction.

3.6 Chapter Summary and Evaluation

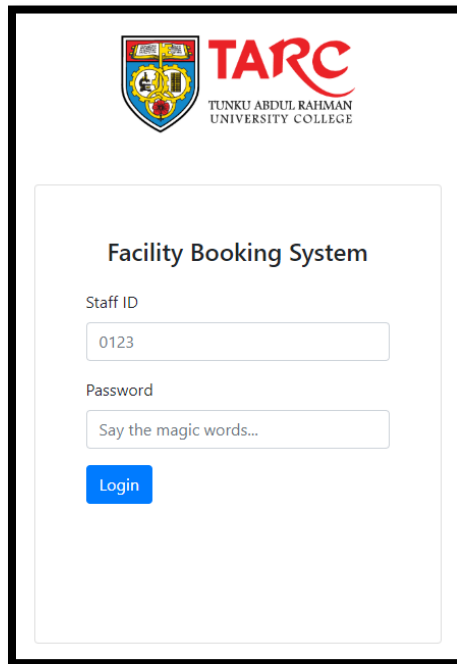
During the preparation of the chapter, choosing a suitable methodology become a challenges because there are so much suitable methodology to be used. At last, prototyping was chosen as out methodology. There are few reason to support the selection, facts are gathered through various method and are recorded. This is because prototyping supports immediate modules changing if there are any modules that are failed to finish or any unforeseen issues are found during any stages of development. This chapter analyses requirements, including functional, non- functional and other external requirements, for this proposed system to determine user expectations for this proposed system.

Chapter 4

System Design

Chapter 4 System Design

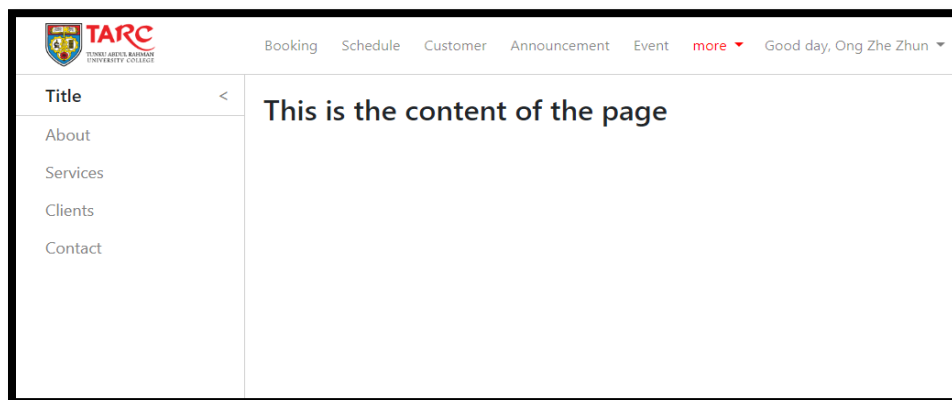
4.1 UI Design



The login page features the TARC logo (Tunku Abdul Rahman University College) at the top left. The main heading is "Facility Booking System". Below this, there are two input fields: "Staff ID" with the value "0123" and "Password" with the placeholder text "Say the magic words...". A blue "Login" button is positioned below the password field.

Figure 4.1.1: Login Page

The login page uses minimalist design which only include username and password field. This is to make the interface simpler to be used and increase user familiarity.



The homepage layout includes a header with the TARC logo and navigation links: "Booking", "Schedule", "Customer", "Announcement", "Event", "more", and "Good day, Ong Zhe Zhun". A sidebar on the left contains a "Title" section with a less-than sign and a list of links: "About", "Services", "Clients", and "Contact". The main content area displays the text "This is the content of the page".

Figure 4.1.2: Homepage

Once staffs login successfully, they will be redirected to the homepage of the system. Staffs able to redirect to other pages via the header section to done their job.

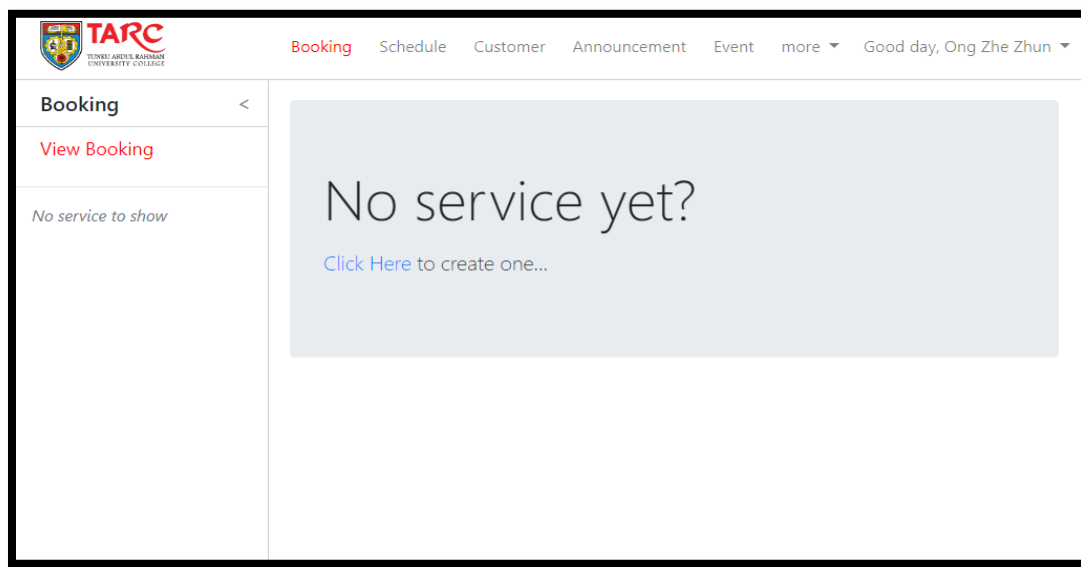


Figure 4.1.3: Booking page with no service available

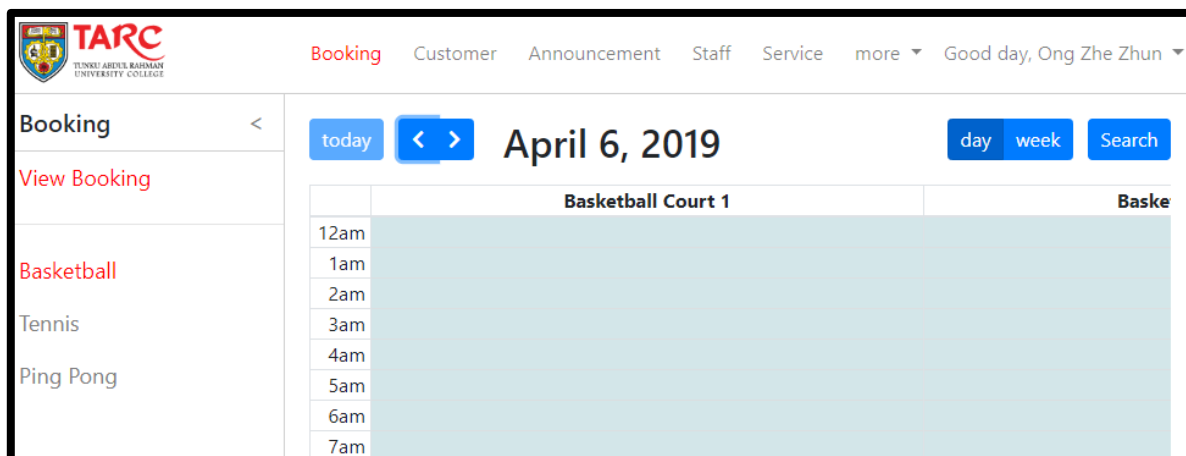


Figure 4.1.4: Booking page if service is available

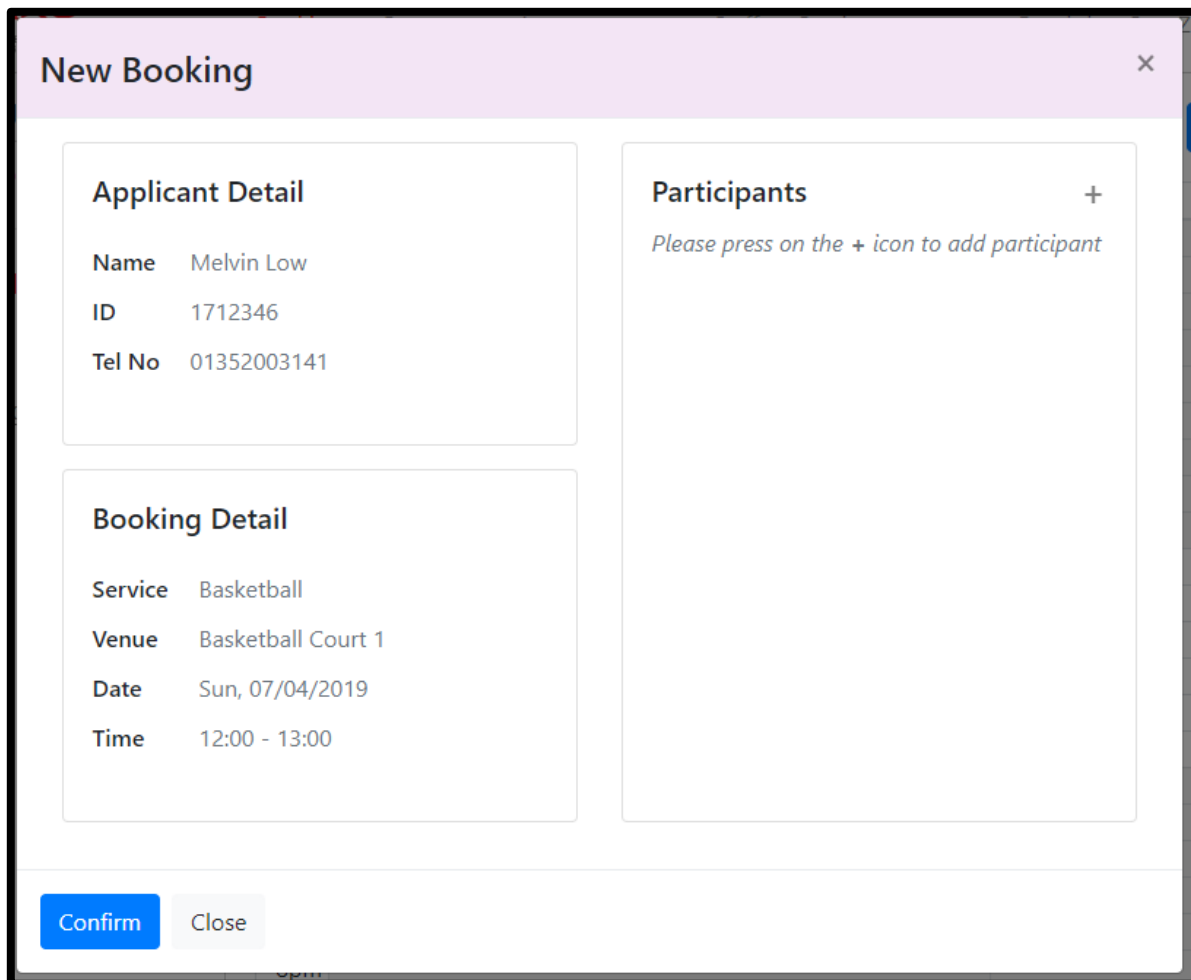
The screenshot shows the TARC Facility Booking System interface. The header includes the TARC logo and navigation links: Booking, Customer, Announcement, Staff, Service, and a dropdown menu. The user is logged in as 'Good day, Ong Zhe Zhun'. The main section displays a calendar for April 6, 2019, with a 'today' button and navigation arrows. A sidebar on the left lists sports: Basketball, Tennis, and Ping Pong. The main calendar area shows a booking for 'Basketball Court 1' from 9:00 to 11:00.

	Basketball Court 1	Baske
12am		
1am		
2am		
3am		
4am		
5am		
6am		
7am		
8am		
9am	9:00 - 11:00	
10am		
11am		

Figure4.1.5: Choosing desired time and venue for booking

The screenshot shows a 'Search Applicant' dialog box. It has a title bar with a close button (X). The main area contains a label 'Reg ID' and a text input field with the value '16WMR12345'. Below the input field are two buttons: 'Search' and 'Close'.

Figure 4.1.6: Entering customer information



The image shows a 'New Booking' form with a purple header bar containing a close button (X). The form is divided into three main sections: 'Applicant Detail', 'Booking Detail', and 'Participants'. The 'Applicant Detail' section contains fields for Name (Melvin Low), ID (1712346), and Tel No (01352003141). The 'Booking Detail' section contains fields for Service (Basketball), Venue (Basketball Court 1), Date (Sun, 07/04/2019), and Time (12:00 - 13:00). The 'Participants' section has a plus icon and a text prompt: 'Please press on the + icon to add participant'. At the bottom of the form are two buttons: 'Confirm' (blue) and 'Close' (grey).

New Booking	
Applicant Detail	
Name	Melvin Low
ID	1712346
Tel No	01352003141
Booking Detail	
Service	Basketball
Venue	Basketball Court 1
Date	Sun, 07/04/2019
Time	12:00 - 13:00
Participants +	
Please press on the + icon to add participant	
Confirm	Close

Figure 4.1.7: Completing booking

The booking module allows staffs to assist customer to perform facility booking. If there are no services available, the system will acknowledge staff to create a new service. Once there are service available, the system will display the service for staff to perform booking action. Staff will first choose the desired time and venue, then entering customer's id to proceed with booking.

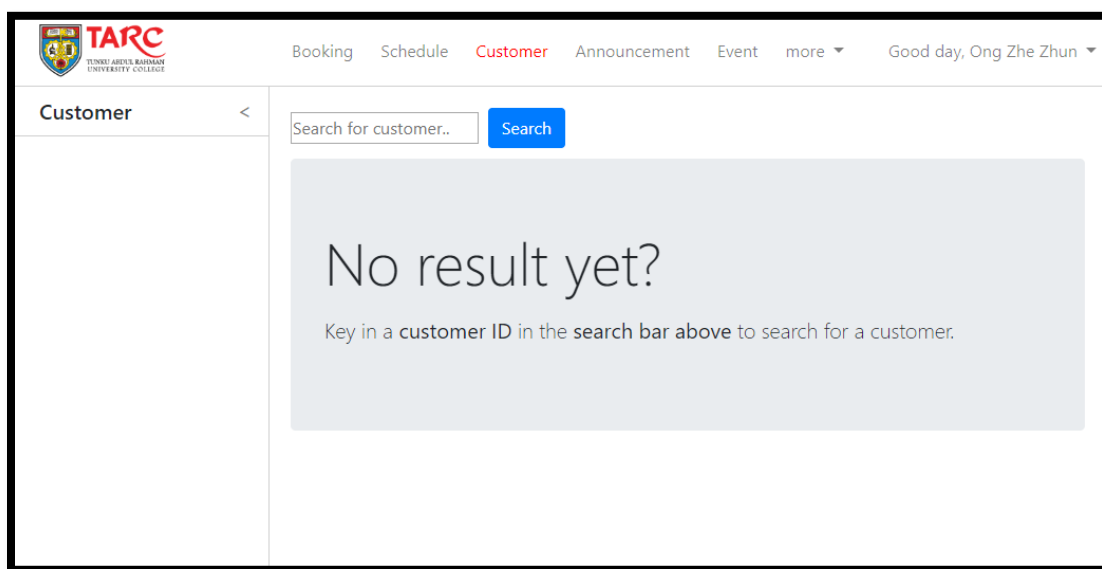


Figure 4.1.8: Customer page

This page is for staff to perform customer related tasks such as viewing their information, processing their registration request, Import customer data from TARUC's database, and perform password recovery.

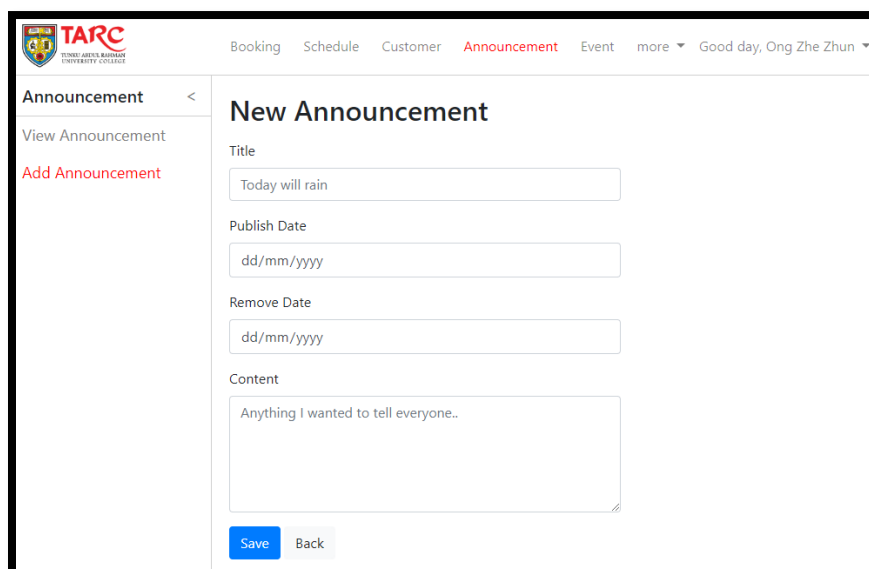


Figure 4.1.9: Create new announcement

This page is for staffs to create new announcement, the created announcement will be sent to customer's mobile app for them to view the announcement.

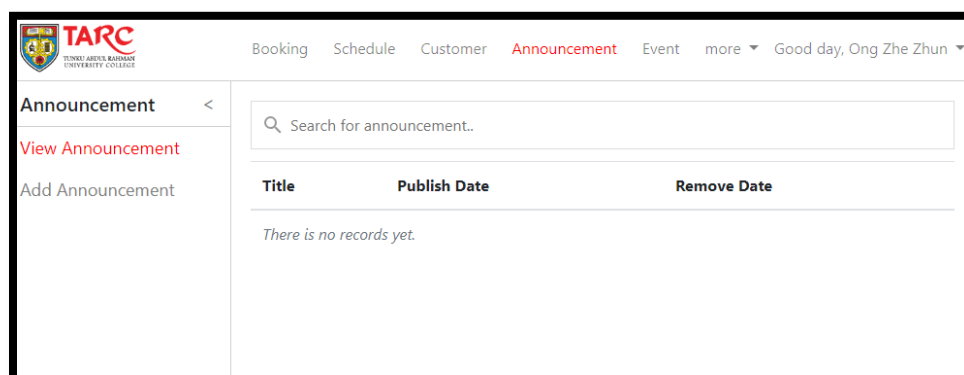


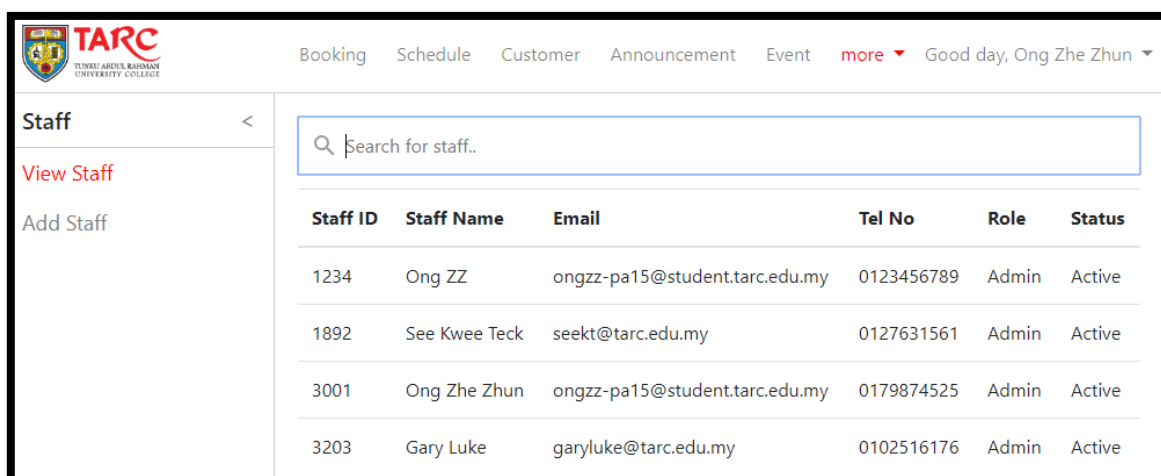
Figure 4.1.10: View announcement

After creating announcement, staff able to view created announcement in this page, editing the information of announcement and even removing the announcement.

The screenshot displays the 'New Staff' form on the TARC website. The top navigation bar is identical to the previous screenshot, with 'Announcement' highlighted. The left sidebar menu for 'Staff' includes links for 'View Staff' and 'Add Staff' (in red). The main form area is titled 'New Staff' and contains several input fields: 'Staff ID' (with 'p123' entered), 'Name' (with 'Adam Teh' entered), 'Email' (with 'adam@example.com' entered), and 'Tel No' (with '0171234567' entered). There is a dropdown menu for 'Role' currently showing '--Please select role'. At the bottom, there are radio buttons for 'Status', with 'active' selected and 'inactive' unselected. 'Save' and 'Back' buttons are located at the bottom of the form.

Figure 4.1.11: Add new staff account

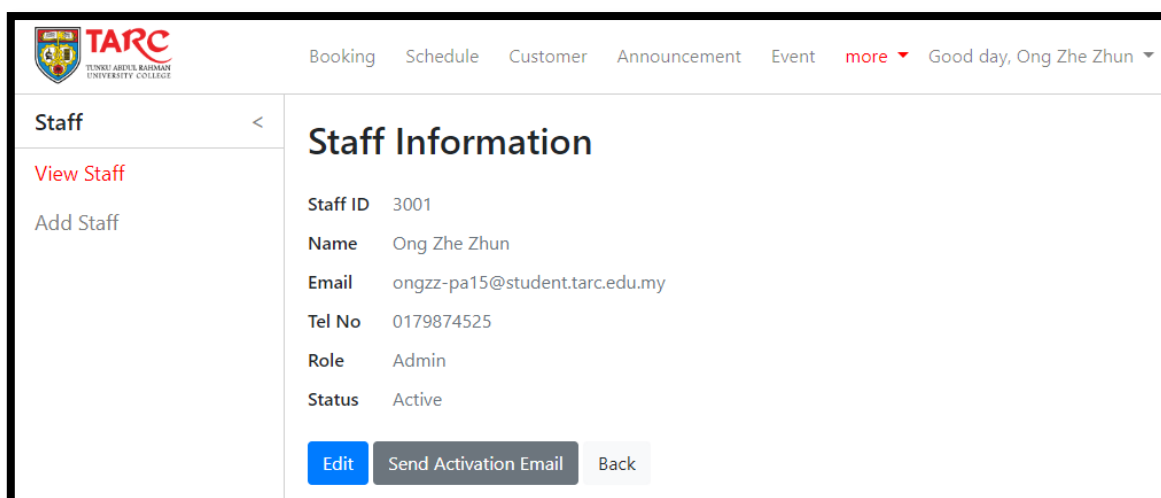
This page is use by staff with higher administrative access to create a new account for other staffs. A verification email will be sent to the provided email address for setting the password and activating the account.



The screenshot shows the TARC Staff Management interface. On the left is a sidebar with the TARC logo and navigation links: Staff, View Staff, and Add Staff. The main area features a search bar and a table of staff accounts.

Staff ID	Staff Name	Email	Tel No	Role	Status
1234	Ong ZZ	ongzz-pa15@student.tarc.edu.my	0123456789	Admin	Active
1892	See Kwee Teck	seekt@tarc.edu.my	0127631561	Admin	Active
3001	Ong Zhe Zhun	ongzz-pa15@student.tarc.edu.my	0179874525	Admin	Active
3203	Gary Luke	garyluke@tarc.edu.my	0102516176	Admin	Active

Figure 4.1.12: View staff account



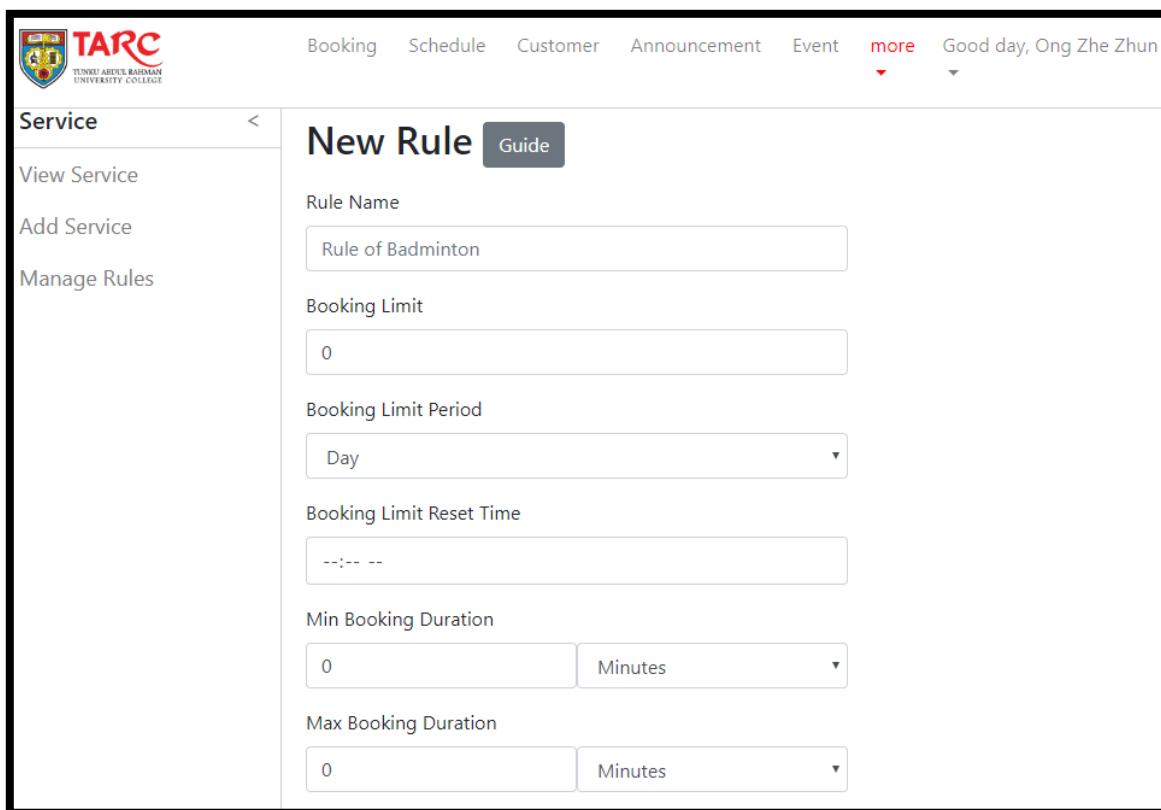
The screenshot shows the TARC Staff Information page. The sidebar is the same as in Figure 4.1.12. The main area displays the details for staff ID 3001, Ong Zhe Zhun, with fields for Name, Email, Tel No, Role, and Status. At the bottom are buttons for Edit, Send Activation Email, and Back.

Staff ID	3001
Name	Ong Zhe Zhun
Email	ongzz-pa15@student.tarc.edu.my
Tel No	0179874525
Role	Admin
Status	Active

[Edit](#) [Send Activation Email](#) [Back](#)

Figure 4.1.13: View staff account information

This page lets staff to view all the staff account created in the system. Staff able to view the account details and edit the account details, performing password recovery.



TARC
TUNKU ABRAHAM RAHMAN
UNIVERSITY COLLEGE

Booking Schedule Customer Announcement Event **more** Good day, Ong Zhe Zhun

Service <

View Service
Add Service
Manage Rules

New Rule [Guide](#)

Rule Name

Booking Limit

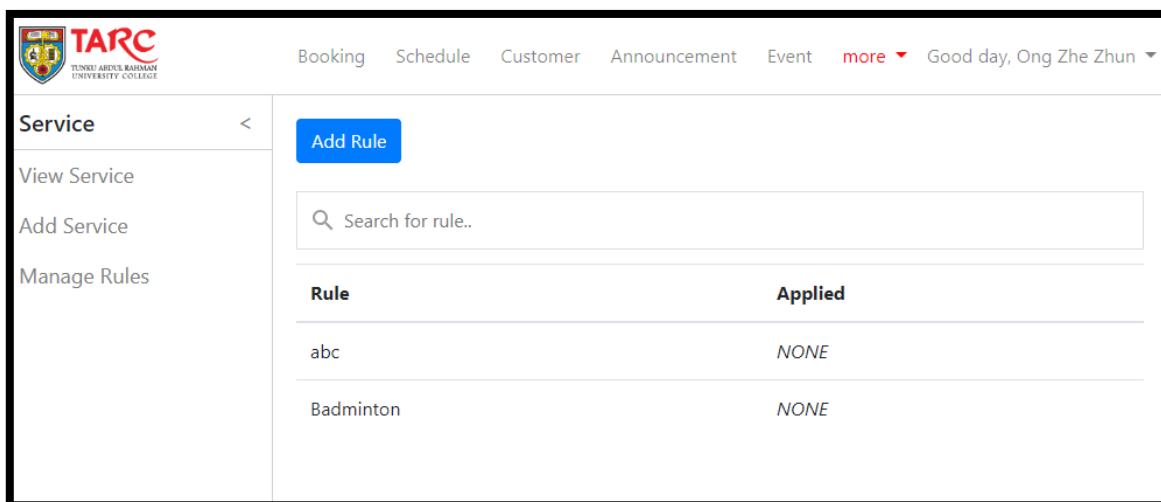
Booking Limit Period

Booking Limit Reset Time

Min Booking Duration

Max Booking Duration

Figure 4.1.14: Create new rule



TARC
TUNKU ABRAHAM RAHMAN
UNIVERSITY COLLEGE

Booking Schedule Customer Announcement Event **more** Good day, Ong Zhe Zhun

Service <

View Service
Add Service
Manage Rules

[Add Rule](#)

Rule	Applied
abc	NONE
Badminton	NONE

Figure 4.1.15: View rules

TARC
TUNJATI ARJUNA RAHMAN
UNIVERSITY COLLEGE

Booking Schedule Customer Announcement Event **more** Good day, Ong Zhe Zhun

Service <

View Service
Add Service
Manage Rules

Update Rule Guide

Rule Name
abc

- The service has booking limit of per which will be reset on .

NOTE: Booking can be made in daily basis and reset on 01:00 am everyday. Example, 14th Nov 2018 01:00 - 15th Nov 2018 01:00 is the period of time your customer be able to make booking and use up their booking limits and quota. 14th Nov 2018 01:00 will be the start of booking, then it will be reset on 15th Nov 2018 01:00 and start a new section.

- The booking duration has a minimum of and a maximum of .
- A booking can be made in advance and the last booking can be made before the reset

Figure 4.1.16: Update rules with guidelines

Figure 4.1.15 – 4.1.17 is the pages meant to creating new facility rules, view created rules, or modifying the rules. Rules must be established by staffs first before they can create a new service to serve the facility available for booking.

TARC
TUNJATI ARJUNA RAHMAN
UNIVERSITY COLLEGE

Booking Schedule Customer Announcement Event **more** Good day, Ong Zhe Zhun

Service <

View Service
Add Service
Manage Rules

New Service

Service ID

Service Name

Booking Rule

Category

Availability

Save Back

Figure 4.17: Create new service

The screenshot shows the 'Service' management interface. On the left, a sidebar contains the 'Service' menu item. The main content area features a search bar labeled 'Search for service..'. Below the search bar, a table is displayed with the following headers: 'Service ID', 'Service Name', 'Category', and 'Type'. The table body is empty, showing the message 'There is no records yet.'

Figure 4.1.18: View service

The service page allows staffs to make the facility booking possible by creating services. Before creating new service, certain requirement are need to be achieved, such as there must at least one rules created. Once the service was created, the facility or resources will be available for customer for booking.

The screenshot shows the 'New Venue' form. The form includes the following fields and options:

- Venue ID:** A text input field containing 'sc'.
- Venue Name:** A text input field containing 'Sport Complex'.
- Operation Hour:** A dropdown menu with 'Standard Hours' selected.
- Status:** Radio buttons for 'Available' (selected), 'Under Maintenance', and 'Closed'.
- Location:** A text input field containing '10.3, 36.365'.

At the bottom of the form, there are two buttons: 'Save' (in blue) and 'Back' (in grey).

Figure 4.1.19: Create new venue

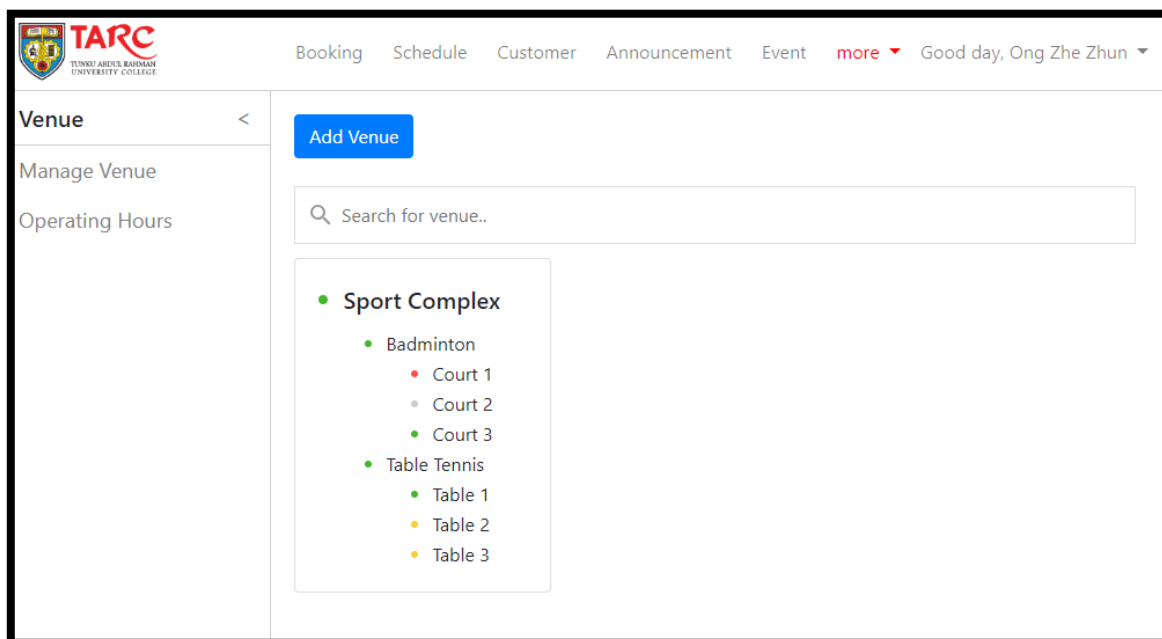


Figure 4.1.20: View venue

The Venue module or pages is used to record all the facility that will be opened for booking. Only facilities or resources that was recorded in the system will be available for booking.

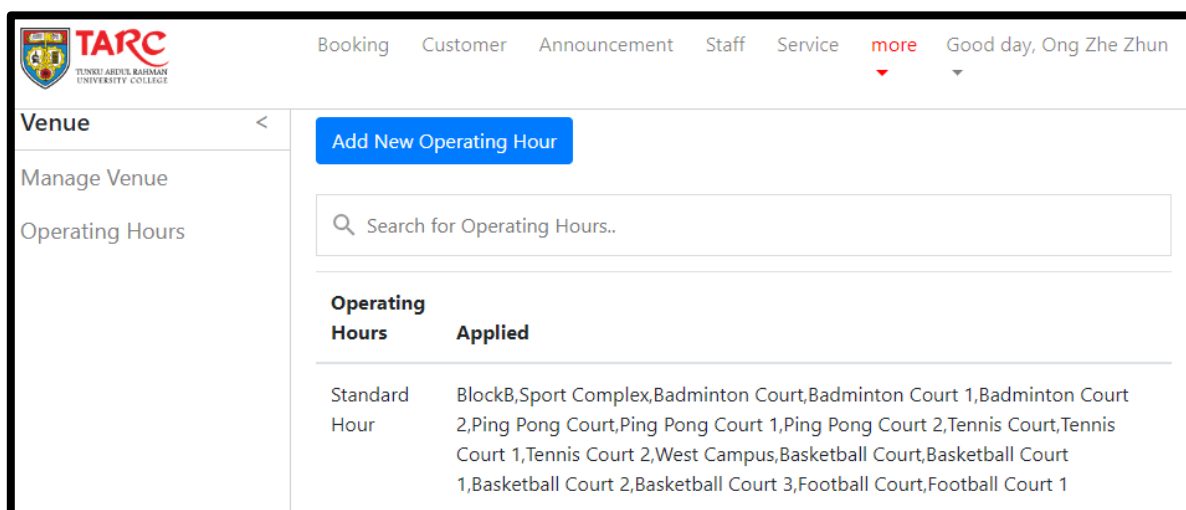


Figure 4.1.21: View operating hour

TARC
TUNKUT ABDUL RAHMAN
UNIVERSITY COLLEGE

Booking Customer Announcement Staff Service more Good day, Ong Zhe Zhun

Venue <

Manage Venue

Operating Hours

Update Operating Hour

Name
Standard Hour

Operating Hours [Add Day](#)

Start Day	End Day	Open Time	Closing Time	
Monday	Friday	09:00 AM	06:00 PM	×
Saturday	Saturday	09:00 AM	04:00 PM	×
Sunday	Sunday	09:00 AM	03:00 PM	×

[Save](#) [Back](#)

Figure 4.1.22: Manage operating hour

The operating hours is used to control the operating hour or bookable hour of the venue for the services. Bookings only able to be made within the operating hour defined.

4.2 Data Design

4.2.1 Entity Relationship Diagram (ERD)

In order to create the Entity Relationship Diagram (ERD), lucidchart was used as the designing tool as it is free yet able to export the diagram in any kind of format for backup purposes. Therefore, changes can be done quickly if any problem was found with the ERD.

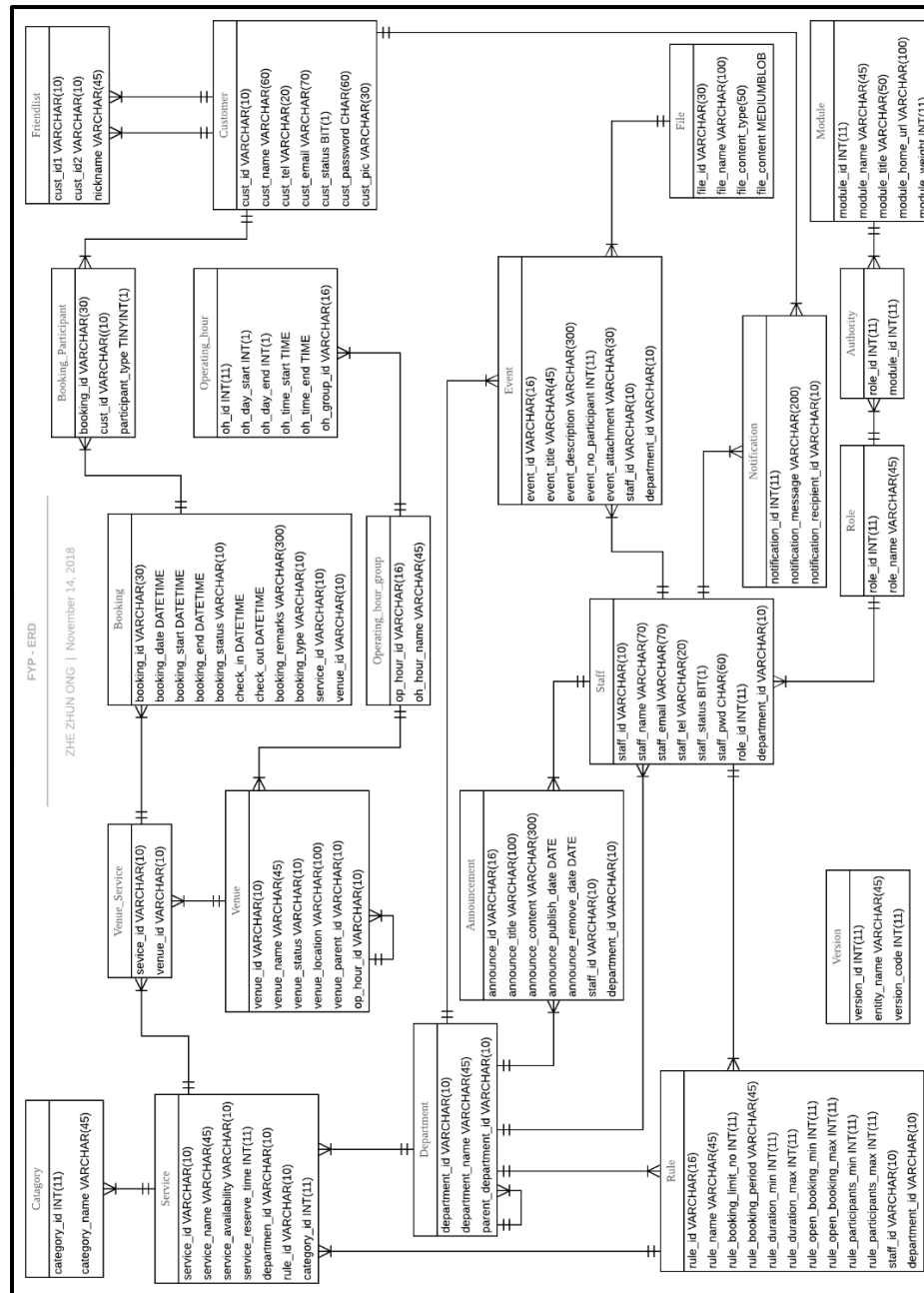


Figure 4.2.1: ERD for facility booking system

4.2.2 Data Dictionary of Tables

Table 4.2.1: Customer Table

Field Name	Data Type	Field Length	Constrain	Description
cust_id	VARCHAR	10	Primary Key	Customer's Identification Number
cust_name	VARCHAR	60	No	Customer's Full Name
cust_tel	VARCHAR	20	No	Customer's Full Phone Number
cust_email	VARCHAR	70	No	Customer's Full Email
cust_status	BIT	1	No	Customer's Full Status
cust_password	CHAR	60	No	Customer's Full Password
cust_pic	VARCHAR	30	No	Customer's Full Profile Picture

Table 4.2.2 Friend List Table

Field Name	Data Type	Field Length	Constrain	Description
cust_id1	VARCHAR	10	Foreign Key	Customer1's Identification Number
cust_id2	VARCHAR	10	Foreign Key	Customer2's Identification Number
nickname	VARCHAR	45	No	Customer2's Nickname

Table 4.2.3 Booking Table

Field Name	Data Type	Field Length	Constrain	Description
booking_id	VARCHAR	30	Primary Key	Booking's Identification Number
booking_date	DATETIME		No	Booking's Date
booking_start	DATETIME		No	Booking's Start Time
booking_end	DATETIME		No	Booking's End Time
booking_status	VARCHAR	10	No	Booking's Status
check_in	DATETIME		No	Booking's Check In Time
check_out	DATETIME		No	Booking's Check Out Time
booking_remarks	VARCHAR	300	No	Booking's Remarks
booking_type	VARCHAR	10	No	Booking's Type
service_id	VARCHAR	10	Foreign Key	Service's Identification Number
venue_id	VARCHAR	10	Foreign Key	Venue's Identification Number

Table 4.2.4 Booking Participant Table

Field Name	Data Type	Field Length	Constrain	Description
booking_id	VARCHAR	30	Foreign Key	Booking's Identification Number
cust_id	VARCHAR	10	Foreign Key	Customer's Identification Number
participant_type	TINYINT	1	No	Participant's type (Lecturer/Student?)

Field Name	Data Type	Field Length	Constrain	Description
venue_id	VARCHAR	10	Primary Key	Venue's Identification Number
venue_name	VARCHAR	45	No	Venue's Name
venue_status	VARCHAR	10	No	Venue's Status
venue_location	VARCHAR	100	No	Venue's Location
venue_parent_id	VARCHAR	10	No	Venue's Parent Id
op_hour_idd	VARCHAR	10	Foreign Key	Operating_Hour's Index Number

Table 4.2.5: Venue Table

Table 4.2.6 Department Table

Field Name	Data Type	Field Length	Constrain	Description
announce_id	VARCHAR	16	Primary Key	Announcement ID
announce_title	VARCHAR	100	No	Announcement's Title
announce_content	VARCHAR	300	No	Announcement's content
announce_publish_date	DATE		No	Announcement publish date
announce_remove_date	DATE		No	Announcement removed date
staff_id	VARCHAR	10	Foreign Key	Staff Identification Number
department_id	VARCHAR	10	Foreign Key	Department Identification Number

Field Name	Data Type	Field Length	Constrain	Description
announce_id	VARCHAR	16	Primary Key	Announcement ID
announce_title	VARCHAR	100	No	Announcement's Title
announce_content	VARCHAR	300	No	Announcement's content
announce_publish_date	DATE		No	Announcement publish date
announce_remove_date	DATE		No	Announcement removed date
staff_id	VARCHAR	10	Foreign Key	Staff Identification Number
department_id	VARCHAR	10	Foreign Key	Department Identification Number

Table 4.2.7 Announcement Table

Table 4.2.8 Event Table

Field Name	Data Type	Field Length	Constrain	Description
event_id	VARCHAR	16	Primary Key	Event Identification Number
event_title	VARCHAR	45	No	Event's title
event_description	VARCHAR	300	No	Event's description
event_no_participant	INT	11	No	No of participant in event
event_attachment	VARCHAR	30	No	Event's attachment
staff_id	VARCHAR	10	Foreign Key	Staff Identification Number
department_id	VARCHAR	10	Foreign Key	Department Identification Number

Table 4.2.9 Staff Table

Field Name	Data Type	Field Length	Constrain	Description
staff_id	VARCHAR	10	Primary Key	Staff's Identification Number
staff_name	VARCHAR	70	No	Staff's name
staff_email	VARCHAR	70	No	Staff's email
staff_tel	VARCHAR	20	No	Staff's telephone number
staff_status	BIT	1	No	Staff's current status
staff_pwd	CHAR	60	No	Staff's login password
role_id	INT	11	Foreign Key	Role's Identification Number
department_id	VARCHAR	10	Foreign Key	Department Identification Number

Table 4.2.10 Rule Table

Field Name	Data Type	Field Length	Constrain	Description
rule_id	VARCHAR	16	Primary Key	Rule's identification Number
rule_name	VARCHAR	45	No	Rule's name
rule_booking_limit_no	INT	11	No	Number limit of participant in one booking
rule_booking_period	VARCHAR	45	No	Booking period
rule_duration_min	INT	11	No	Minimum time duration in one booking
rule_duration_max	INT	11	No	Maximum time duration in one booking
rule_open_booking_min	INT	11	No	Minimum number of booking open for book

rule_open_booking_max	INT	11	No	Maximum number of booking open for book
staff_id	VARCHAR	10	Foreign Key	Staff's identification Number
department_id	VARCHAR	10	Foreign Key	Department's identification Number

Table 4.2.11 Service Table

Field Name	Data Type	Field Length	Constrain	Description
service_id	VARCHAR	10	Primary Key	Service's Identification Number
service_name	VARCHAR	45	No	Service's name
service_availability	VARCHAR	10	No	Service's availability
service_reserve_time	INT	11	No	Service reserve's time
department_id	VARCHAR	10	Foreign Key	Department's Identification Number
rule_id	VARCHAR	10	Foreign Key	Rule's Identification Number
category_id	INT	11	Foreign Key	Category's Identification Number

Table 4.2.12 Department Table

Field Name	Data Type	Field Length	Constrain	Description
department_id	VARCHAR	10	Primary Key	Department's Identification Number
department_name	VARCHAR	45	No	Department's name
parent_department_id	VARCHAR	10	Foreign Key	Parent of department's identification number

Table 4.2.13 Operating Hour Table

Field Name	Data Type	Field Length	Constrain	Description
oh_id	INT	11	Primary Key	Operating hour Identification Number
oh_day_start	INT	1	No	Operating start of the day
oh_day_end	INT	1	No	Operating end of the day
oh_time_start	TIME		No	Operating start of the time
oh_time_end	TIME		No	Operating end of the time
oh_group_id	VARCHAR	16	Foreign Key	Operating group identification number

Table 4.2.14 Operating Hour Group Table

Field Name	Data Type	Field Length	Constrain	Description
op_hour_id	VARCHAR	16	Primary Key	Operation hour group identification number
oh_hour_name	VARCHAR	45	No	Operation hour group name

Table 4.2.15 Venue Service Table

Field Name	Data Type	Field Length	Constrain	Description
service_id	VARCHAR	10	Foreign Key	Service's Identification Number
venue_id	VARCHAR	10	Foreign Key	Venue's Identification Number

Table 4.2.16 Category Table

Field Name	Data Type	Field Length	Constrain	Description
category_id	INT	11	Primary Key	Category's Identification Number
category_name	VARCHAR	45	No	Category's name

Table 4.2.17 Version Table

Field Name	Data Type	Field Length	Constrain	Description
version_id	INT	11	Primary Key	Version's Identification Number
entity_name	VARCHAR	45	No	entity name of version
version_code	INT	11	No	System's version number

Table 4.2.18 File Table

Field Name	Data Type	Field Length	Constrain	Description
file_id	VARCHAR	30	Primary key	File's Identification Number
file_name	VARCHAR	100	No	File's name
file_content_type	VARCHAR	50	No	File's content type
file_content	MEDIUMBLOB		No	File's content

Table 4.2.19 Notification Table

Field Name	Data Type	Field Length	Constrain	Description
notification_id	INT	11	Primary key	Notification Identification Number
notification_message	VARCHAR	200	No	Notification message
notification_recipient_id	VARCHAR	10	Foreign Key	Customer's ID

Table 4.2.20 Role Table

Field Name	Data Type	Field Length	Constrain	Description
role_id	INT	11	Primary Key	Role's Identification Number
role_name	VARCHAR	45	No	Role's Name

Table 4.2.21 Authority Table

Field Name	Data Type	Field Length	Constrain	Description
role_id	INT	11	Primary Key	Role's Identification Number
module_id	INT	11	Foreign Key	Module's Identification Number

Table 4.2.22 Module Table

Field Name	Data Type	Field Length	Constrain	Description
module_id	INT	11	Primary Key	Module's Identification Number
module_name	VARCHAR	45	No	Module's name
module_file	VARCHAR	50	No	
module_home_url	VARCHAR	100	No	Url to the module
module_weight	INT	11	No	

4.3 Security Design

1. Login

The login UI will show the form which contain username field and password field. The length of username consists of 7 characters that are numbers, while the length of password consists of maximum 7 characters which is the format of the student ID. During typing the password in the password field, the character will be shown as “*****” to maintain the security issue. The username will be use the student ID and the initial password will use the student’s IC number as original password, user can change password in the system. Before the user can access the system, the user need to input the correct username and password.

2. Password Recovery

If the user forgot the password, the user can use the password recovery to solve the issue. The password recovery can be found at the bottom of password field which label with “Forgot password”. When the user clicks the hypertext link, the page of password recovery will be opened. The user need to insert the necessary information such as email address, if the information inserted correctly, the system will allow the send an email to user’s email address, then user can perform password recovery task by accessing the email link that system has sent.

3. Admin Role Management

The design of this admin role management is to control what a user can control by restraint them to use feature that is not supposed to use by certain type of user. By implement this, admin can configure what type of user can access or control which features of the system which provide a lot of security and to prevent certain user abuse the system or use the feature the specific user not supposed to. For example, head of DSA department has full access to the system. Then, the staff of DSA can only control the feature of students that provide the access to configure student details and etc. And also, the staff of Sport Complex can only control the feature of court booking features to help students performing facility booking. There would be a list of staffs where user can choose which staff could access what type of features.

4.4 Process Design

Use Case Diagram

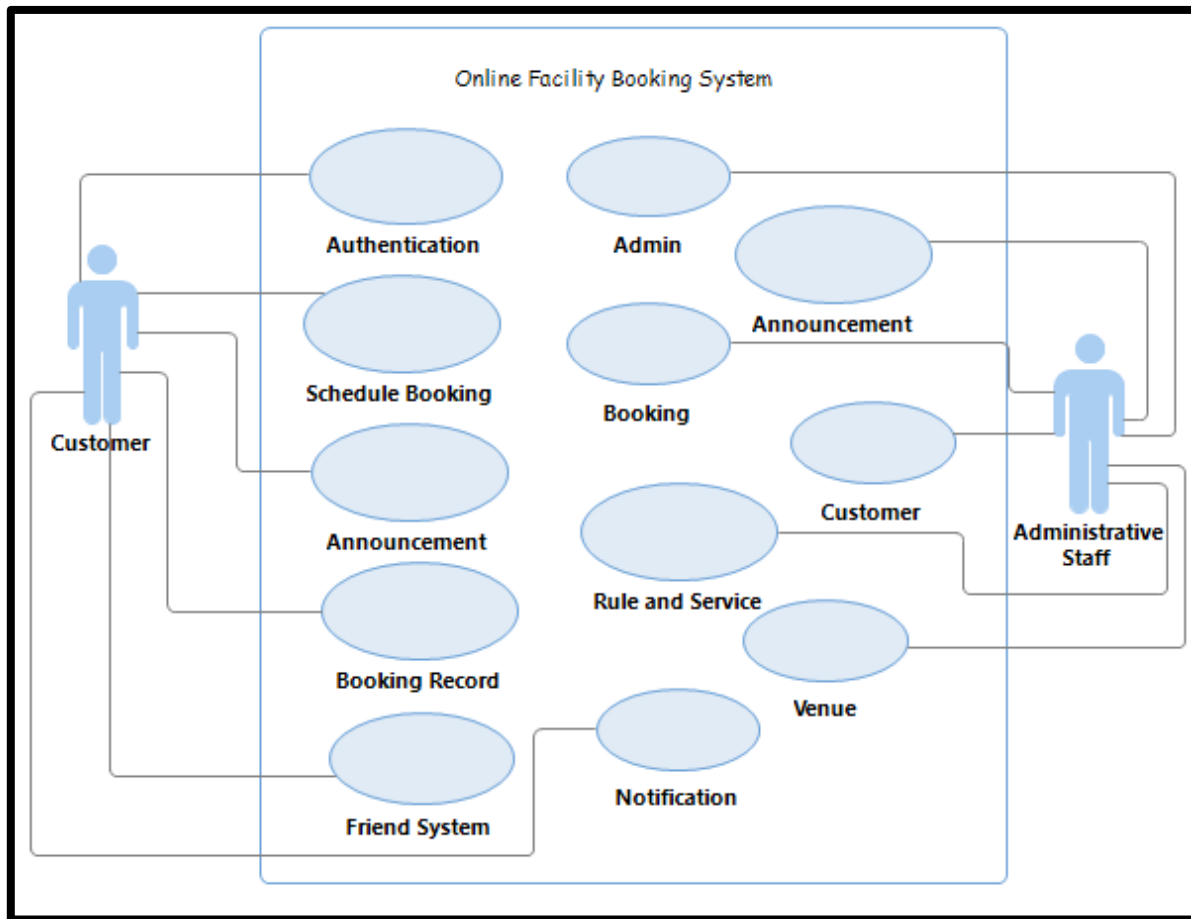


Figure 4.4.1: Overview Use Case Diagram of Online Facility Booking System

The module that currently handled and maintained are:

- a. Admin
- b. Announcement
- c. Booking
- d. Customer
- e. Rule and Service
- f. Venue

Other modules can be refer to Melvin Low Eik Kit's chapter 4 report.

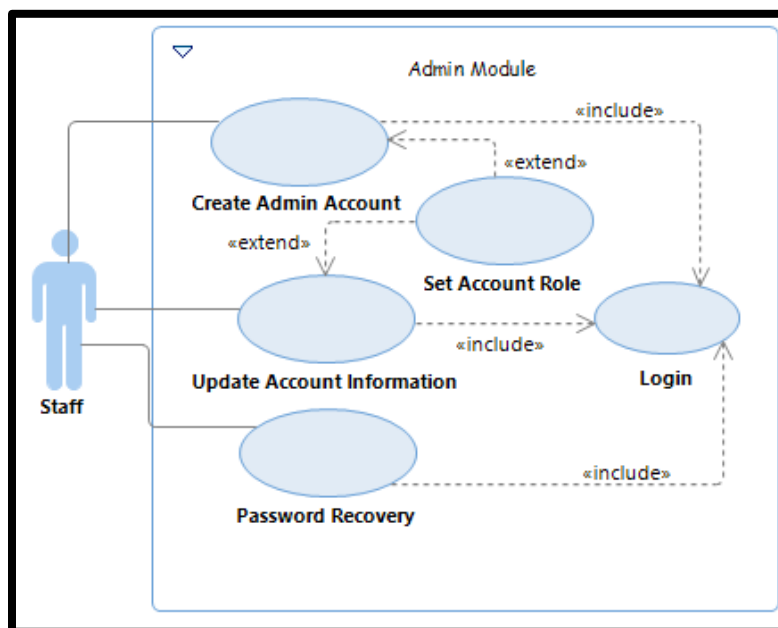


Figure 4.4.2: Use Case Diagram for Admin Module

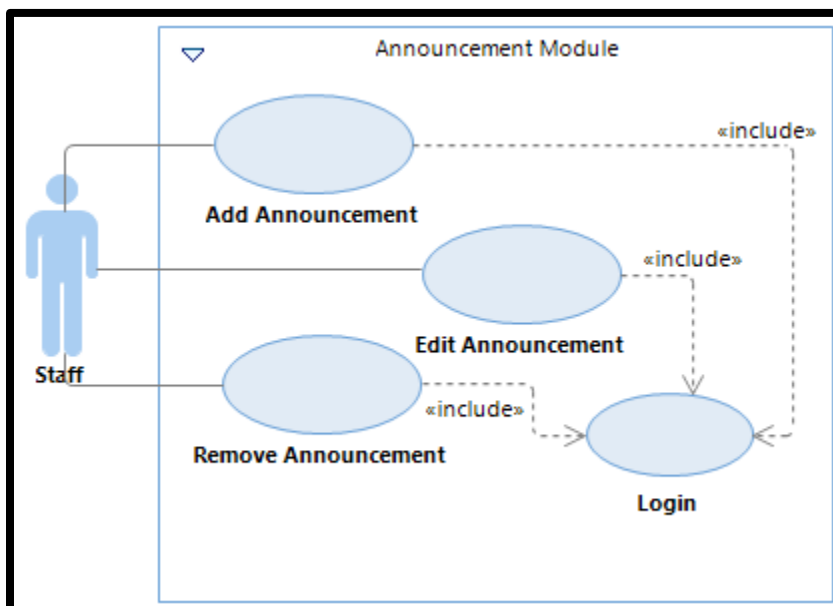


Figure 4.4.3: Use Case Diagram for Announcement Module

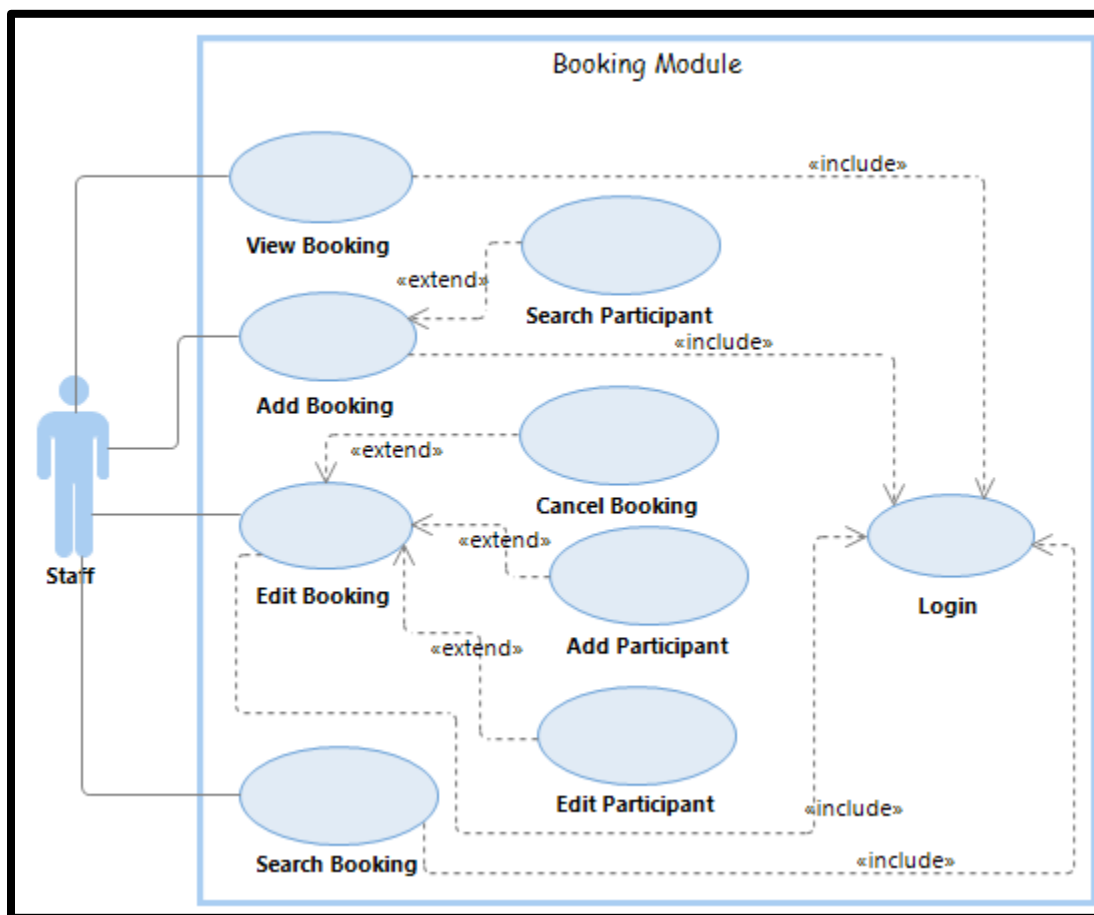


Figure 4.4.4: Use Case Diagram for Booking Module

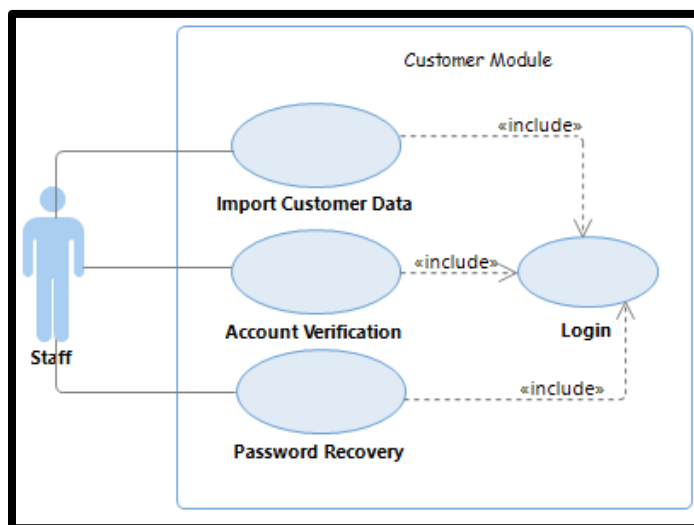


Figure 4.4.4: Use Case Diagram for Customer Module

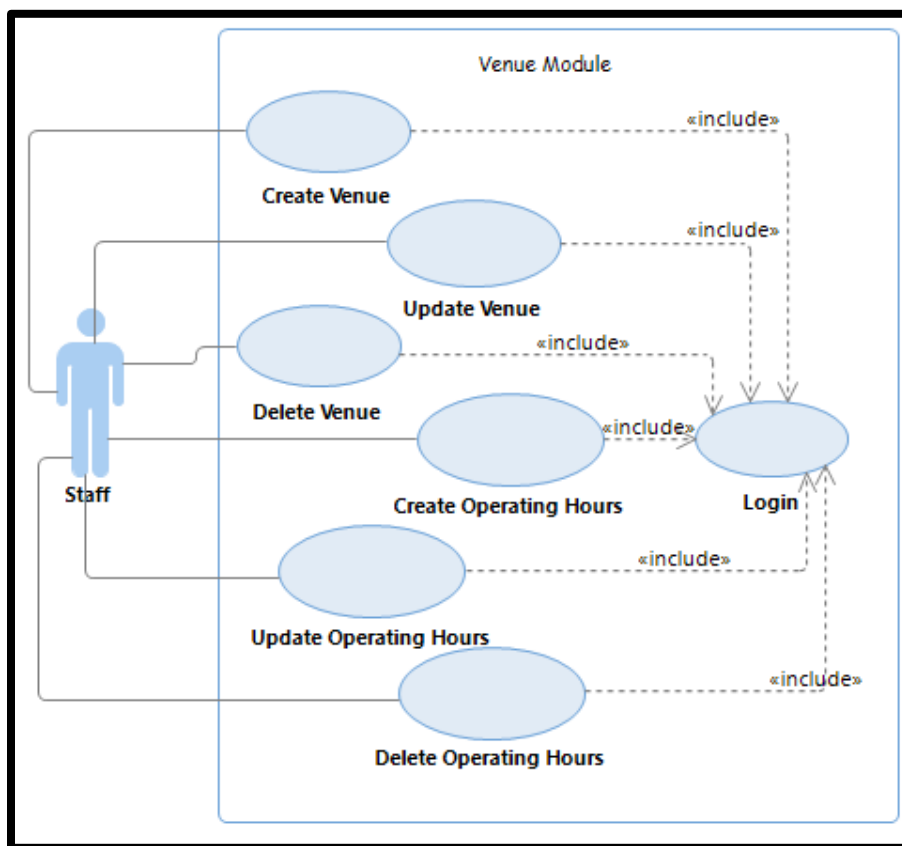


Figure 4.4.5: Use Case Diagram for Venue Module

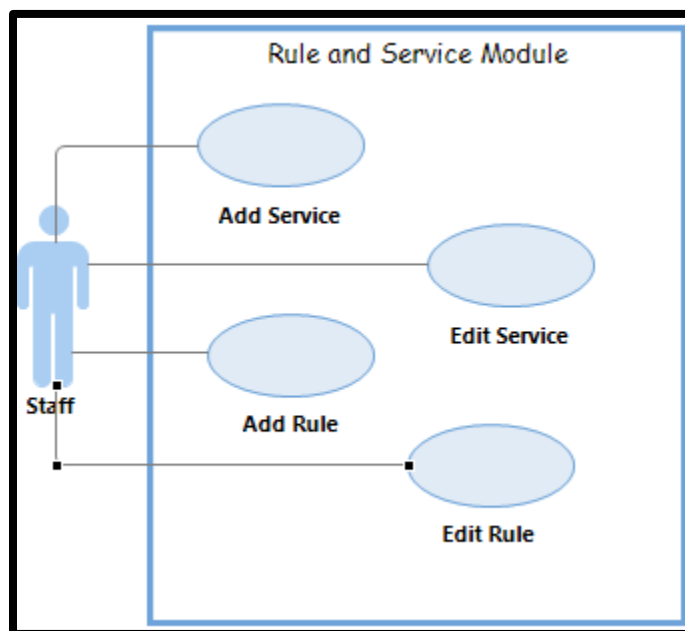


Figure 4.4.6 Use Case Diagram for Rule and Service Module

Use Case Description

Table 4.4.1: Use Case Description for Create Admin Account

Use Case Name	Create Admin Account
Actor	Staff
Brief Description	This use case allow staff with higher accessibility to create an admin account for other staffs
Main Flow of Activities:	
Actor Action	System Response
	1. Display Admin UI
2. Choose "Add Staff" selection in Admin UI	3. Display the Add Staff UI
4. Fill in all the account information and click "create"	5. Display create successful message
Alternative Flow of Activities:	
Step 4: If the information was incorrect, prompt error message to notice the staff to insert the valid information	

Table 4.4.2 Use Case Description for Update Account Information

Use Case Name	Update Account Information
Actor	Staff
Brief Description	This use case allow staff with higher accessibility to update his/her own account information or other staff account's
Main Flow of Activities:	
Actor Action	System Response
	1. Display Admin UI
2. Choose "View Staff" selection in Admin UI	3. Display the View Staff UI
4. Select on an account desired to change	5. Redirect and Display selected account's information
6. Select "Edit" selection	7. Unlock account's information for staff to

	edit
7. Select "Save" after finish Editing	8. Display confirmation box
9. Confirm the editing	10. Display edit successful message
Alternative Flow of Activities:	
Step 7: If the information was incorrect, prompt error message to notice the staff to insert the valid information	

Table 4.4.3 Use Case Description for Import Customer Data

Use Case Name	Import Customer Data
Actor	Staff
Brief Description	This use case allow staff with accessibility to customer module to import customer data from external sources
Main Flow of Activities:	
Actor Action	System Response
	1. Display Customer UI
2. Choose "Import Customer Data" selection in Customer UI	3. Display the Import UI
4. Search for data files to be import and select "import"	5. Import the data file into database
	6. Display import successful message
Alternative Flow of Activities:	
Step 4: If the import was failed, prompt failure message and let staff to re-import	

Table 4.4.4 Use Case Description for Account Verification

Use Case Name	Account Verification
Actor	Staff
Brief Description	This use case allow staff with accessibility to customer module to process customer's account registration request
Main Flow of Activities:	
Actor Action	System Response
	1. Display Customer UI
2. Choose "Account Verification" selection in Customer UI	3. Display the Verification UI
4. select on one of the pending register request	5. Display the selected request information
6. Accept the account register request	7. Display accepted message
	8. Send register successful message to the customer account
Alternative Flow of Activities:	
Step 4: If the import was failed, prompt failure message and let staff to re-import	

Table 4.4.5 Use Case Description for Add Service

Use Case Name	Add Service
Actor	Staff
Brief Description	This use case allow staff add a new service
Main Flow of Activities:	
Actor Action	System Response
	1. Display Service UI
2. Choose "Add Service " selection in Service UI	3. Display the Add Service UI
4. Fill in all the service information and click "create"	5. Display create successful message

Alternative Flow of Activities:	
Step 4: If the information was incorrect, prompt error message to notice the staff to reinsert the valid information	

Table 4.4.6 Use Case Description for Update Service

Use Case Name	Update Service
Actor	Staff
Brief Description	This use case allow staff to modify the information of a service
Main Flow of Activities:	
Actor Action	System Response
	1. Display Service UI
2. Choose "View Service " selection in Service UI	3. Display the View Service UI
4. Select on an service desired to change	5. Redirect and Display selected service's information
6. Select "Edit" selection	7. Allow staff to perform information editing
8. Select "Save" after finish editing	9. Display confirmation box
9. Confirm the editing	10. Display edit successful message
Alternative Flow of Activities:	
Step 8: If the information was incorrect, prompt error message to notice the staff to insert the valid information	

Table 4.4.7 Use Case Description for Create Announcement

Use Case Name	Create Announcement
Actor	Staff
Brief Description	This use case allow staff to create new announcement
Main Flow of Activities:	
Actor Action	System Response
	1. Display Announcement UI
2. Choose "Add Announcement" selection in Announcement UI	3. Display the Add Announcement UI
4. Fill in announcement information and confirm the creation	5. Display create successful message
Alternative Flow of Activities:	
Step 4: If the information was invalid prompt failure message and let staff to re-enter	

Table 4.4.8 Use Case Description for Create Venue

Use Case Name	Create Venue
Actor	Staff
Brief Description	This use case allow staff to add new venue
Main Flow of Activities:	
Actor Action	System Response
	1. Display Venue UI
2. Choose "Add Venue" selection in Venue UI	3. Display the Add Venue UI
4. Fill in all the venue information and click "create"	5. Display create successful message
Alternative Flow of Activities:	
Step 4: If the information was incorrect, prompt error message to notice the staff to reinsert the valid information	

Activity Diagram

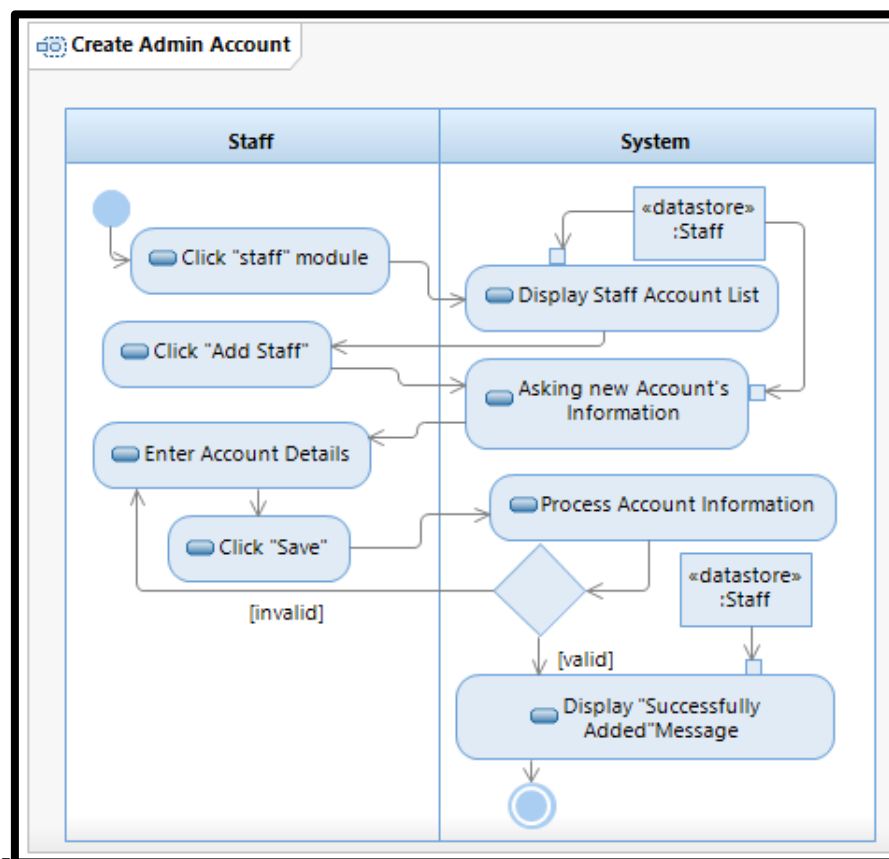


Figure 4.4.7 Activity diagram for create admin account

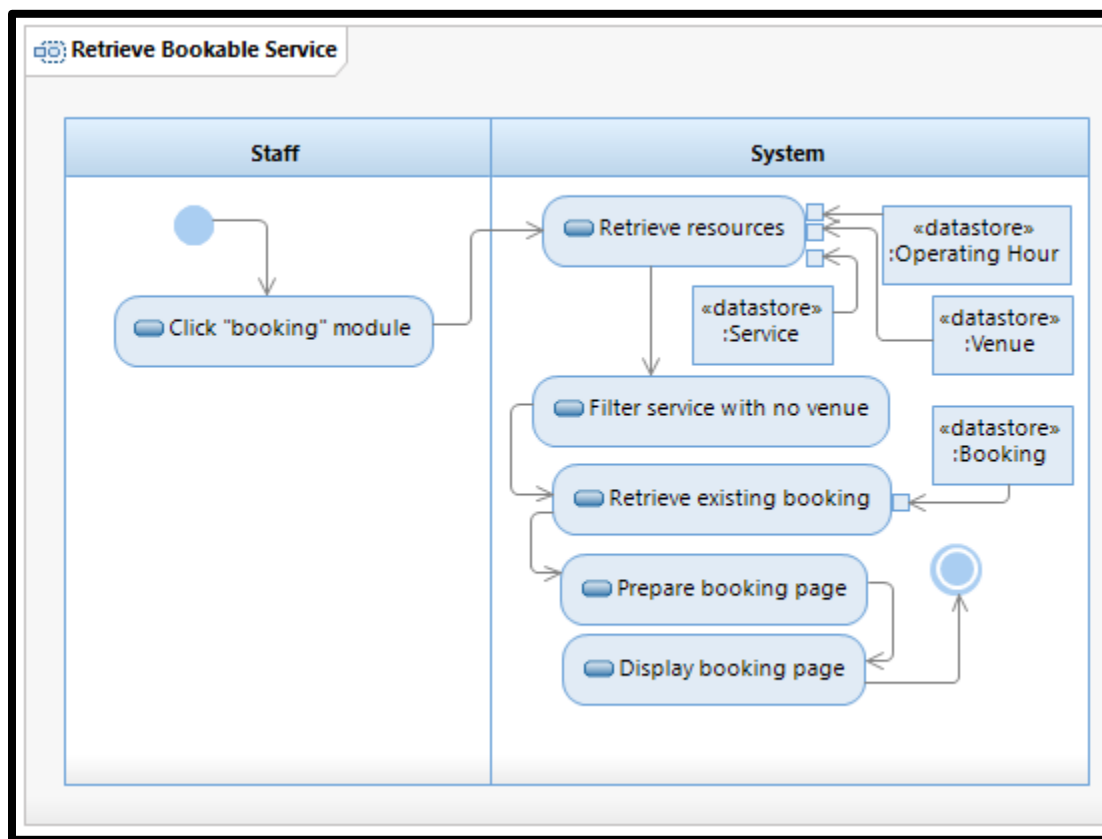


Figure 4.4.8 Activity diagram for retrieve bookable service.

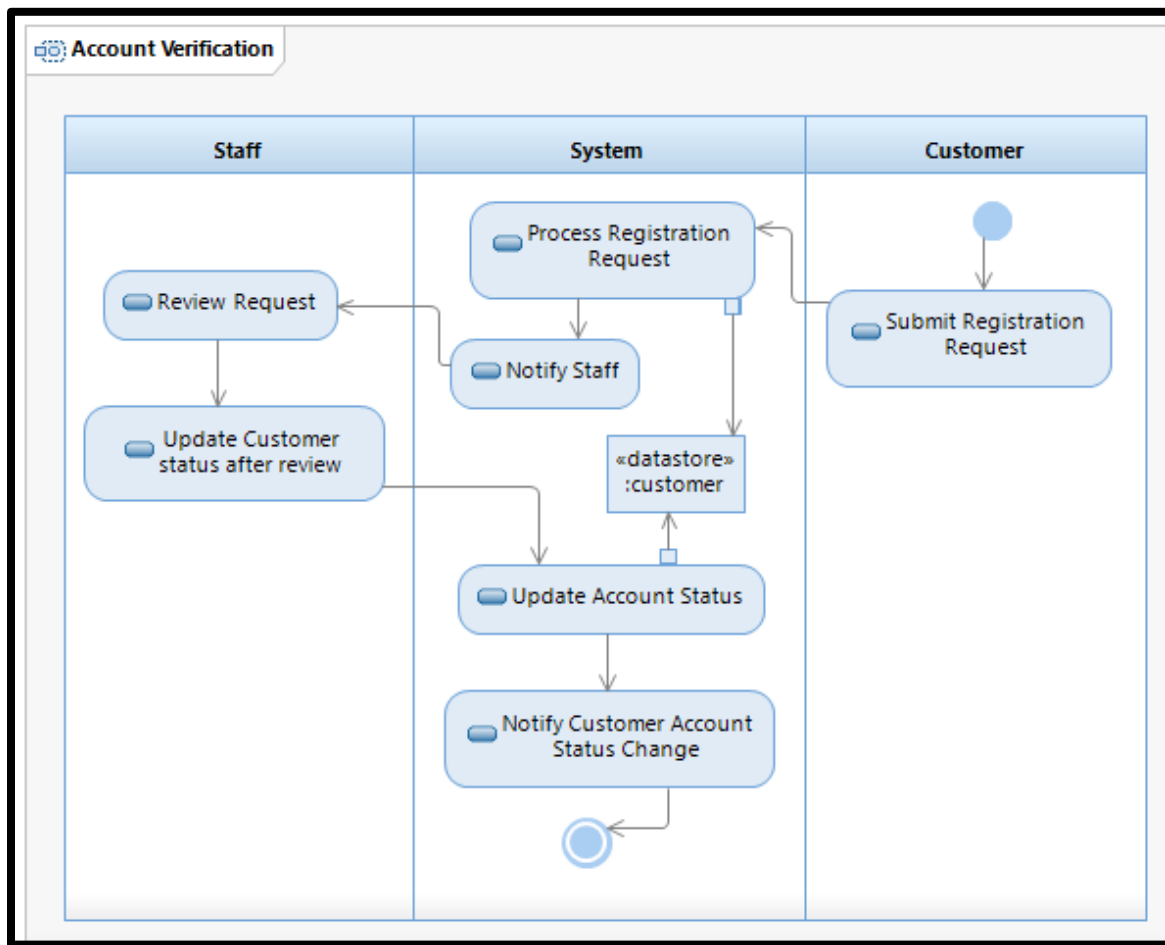


Figure 4.4.9 Activity diagram for account verification.

4.5 Chapter Summary and Evaluation

During the preparation of chapter 4, most of the time was spent on drawing the diagrams and thinking on how the system should perform and how the modules communicate with user. Yet also thinking about the best software to be used to draw all those diagrams.

A software called IBM Rational Software Architecture for WebSphere Software (IBM RSA) was taught by lecturer last year. It is a very useful software that provide lots of convenience in drawing diagrams easy and simple. It just a few simple way such as drag and drop of objects into the diagram and it was done. At first, an online drawing tool called LucidChart was used as replacement of IBM RSA as some issue was faced during the installation of the software.

Throughout the whole development cycle, multiple module changes was faced and different versions of document are needed to be prepare.

In this chapter, most of diagram that able to contribute towards the development of system was included, such as ui design, entity relationship diagram (ERD), use case diagram, activity diagram and security design.

Chapter 5

Testing and Results

Chapter 5 Testing and Results

The Facility Booking System aims to be a real-time booking system that serves all the TARUC's staffs, lecturers, students and alumni. To ensure a smooth and errorless user experience, software testing will be conducted to make sure all possible error checking is available and no error will be produced will the system is deployed.

5.1 Software Testing Methodologies

Software testing is a process to evaluate a software application's functionality in order to determine whether or not the software developed meets the specified requirements and to identify the defects to ensure that the product is free of defects in order to produce a quality product (What Is Software Testing - Definition, Types, Methods, Approaches, 2019). There are numerous types of software testing methods available in the software testing "community". Several prominent software testing methods are used to test the system, mainly white-box testing and black-box testing. Usually, white-box testing will be conducted by software developers, testing on the unit level of the system based on the internal code structure of applications. The test cast are designed using both internal system perspective and programming skills. Black-box testing is a testing method in which testers evaluate the software's functionality without examining the internal code structure of a system. Also, black-box testing mostly used by testers with no programming knowledge, they might have different kind of expectation and approach towards the system.

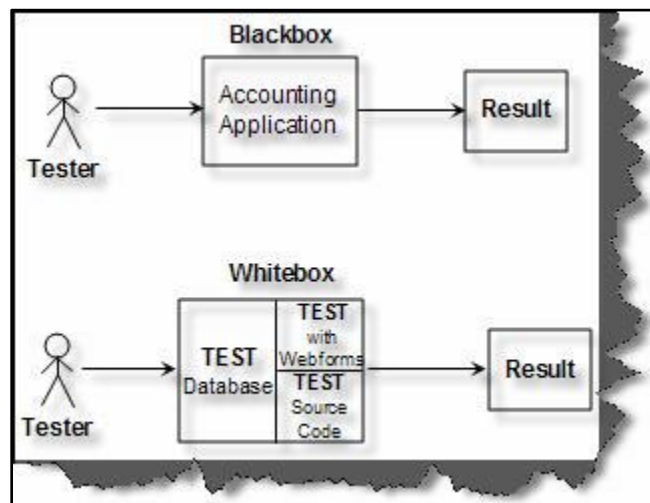


Figure 5.1.1: Difference between Black-Box Testing and White-Box Testing

In this project, black-box testing is conducted, mainly targeting the TARUC staffs working in club house that in charge of all the sport facility booking. Since the staffs might have no IT related knowledge, their opinions and suggestions are very valuable. Their opinions and suggestions definitely will help in further improving the user experience of using the system, also might help in discover extra faults or enhancement of the system.

5.1.1 Results

Test Case

Test case is used during the black-box testing and white-box testing to evaluate the functionality of facility booking system. Test case was used because it is widely used in any system testing phase. The tester able to know what testing is conducting and what kind of result are to be expected, so that tester able to know whether the system is working as expected in the test case. If the system behaves differently, tester able to take note as remarks to express the unexpected behavior of the system. Even during the process developing the test case, it also helps in discovering problems in the requirements or design of the system.

5.2 Test Plan/ Test Case

Table 5.1: Project Details

Project Details			
Student's Name	ONG ZHE ZHUN	Programme:	RSD 3
Project Title	Facility Booking System (Back-end/ server side)	Test Case No.	FBS1000
Module:	Announcement, Admin, Customer and Staff Module	Reference No. (Diagram)	-
Actor(s)	Ong Zhe Zhun (Student)		
Pre-requisites:	1. Staff has an valid account to access Facility Booking System		
Dependencies:	-		

Table 5.2: Test Case

ID	Test Description	Test Action/Inputs	Expected Results	Actual Results	Pass(P)/ Fail(F)	Remarks
FBS 1000	User create or edit an announcement by entering the announcement's title, content, publish date and remove date.	User did not enter information in title and content field.	Prompt message "Please key in content" to notify user to fill in content for the Title	Prompt message "Please key in content" to notify user to fill in content for the Title	P	

			and Content.	and Content.		
FBS 1001	User create or edit an announcement by entering the announcement's title, content, publish date and remove date.	User select a date before current date at publish date field	Prompt message "Please select a date of today or future" to notify user to select correct date.	Prompt message "Please select a date of today or future" to notify user to select correct date	P	
FBS 1002	User create or edit an announcement by entering the announcement's title, content, publish date and remove date.	User select a date before publish date at remove date field	Prompt message "Please select a future date" to notify user to select correct date.	Prompt message "Please select a future date" to notify user to select correct date.	P	
FBS 1003	User create or edit an announcement by entering the announcement's title, content, publish date and remove date.	User select same date for both publish date and remove date field.	Prompt message "Publish Date cannot be same as the Remove Date and Remove Date should be after Publish Date" for either publish date and remove date field to notify user to change date.	Prompt message "Publish Date cannot be same as the Remove Date and Remove Date should be after Publish Date" for either publish date and remove date field to notify user to change date.	P	
FBS 1004	User create or edit an announcement by entering the announcement's title,	User attempt to fail to update a	Update will not occur and the invalid field	Update will not occur and the invalid field	P	

	content, publish date and remove date.	announcement	will surrounded with red indicator to show the field that needed to enter the correct data	will surrounded with red indicator to show the field that needed to enter the correct data		
FBS 1005	User attempt to remove an announcement.	User click on “Remove Announcement”	Update the remove date of the announcement to current date.	Update the remove date of the announcement to current date.	P	
FBS 1006	User attempt to upload a json file containing list of customer information to import into the database.	User attempt to upload nothing to the server.	Prompt error message “Please provide a file containing the customer record”.	Prompt error message “Please provide a file containing the customer record”.	P	
FBS 1007	User attempt to upload a json file containing list of customer information to import into the database.	User attempt to upload a valid json file to the server with valid customer information inside	Prompt successful message “Customer account record imported successfully”.	Prompt successful message “Customer account record imported successfully”.	P	
FBS 1008	User attempt to reset customer’s account password.	User click on “Reset Account Password” button in the customer detail page.	Prompt update successful message, a new password assigned to the account, sent an email containing the	Prompt update successful message, a new password assigned to the account, sent an email containing the	P	

			password to the customer's mailbox.	password to the customer's mailbox.		
FBS 1009	User attempt to update a customer's record.	User left some fillable field empty and submit the update.	System do not perform update, indicating the invalid field with red border and error message at the bottom of the field.	System do not perform update, indicating the invalid field with red border and error message at the bottom of the field.	P	
FBS 1010	User attempt to reset account password via login screen by clicking on "Forget Password?" button.	User entering empty or invalid email.	Prompt invalid message "Error, email might be incorrect!".	Prompt invalid message "Error, email might be incorrect!".	P	
FBS 1011	User update a specific staff's account email.	User enter an invalid format of email	Prompt message "Please fill in with the email format" to notify user to fill in correct email.	Prompt message "Please fill in with the email format" to notify user to fill in correct email.	P	

5.3 Chapter Summary and Evaluation

Software testing plays an extremely important role in the development cycle of a system as it is used to identify whether the system achieved the requirement being made in the planning phase or requested by clients, stakeholders, programmers and even testers that testing on the system.

There are multiple types of software testing methodologies available in the community, and black-box testing and white-box testing are chosen as these two methodologies are the most common. Black-box testing is mostly conducted by software testers by not allowing them to view the source code of the system, letting testers to test the system from their point of view and expectations towards the system. White-box testing is used by software developers throughout the whole development cycle to ensure all the source code and functions they made are functional and have no flaws.

In order to achieve a smooth software testing process, a list of test cases must be well prepared for testers to refer as they probably have zero knowledge regarding the system. All the test cases should be written in simple yet short words to allow testers to understand what the test case is testing on.

Long story short, the system is tested by multiple testers for multiple times. The response shows that the system achieved the functional requirements that was proposed. But there are also some flaws and limitations found in the system that affect its performance. Future enhancement is needed to refine the system to be more reliable.

Chapter 6

Discussions and Conclusion

Chapter 6 Discussion and Conclusion

6.1 Summary

The aim of this project is to develop a new platform that allows student to perform facility booking and staff to manage announcement, admin account, booking, booking or facility resources, and enhance the system with additional features and flaws fixing. The proposed solutions let staffs to have better control on announcement management, customer and staff information, and resource planning (venue and venue's operating hour). Besides, the proposed solution also resolve most of the flaws that was left by previous developers and apply a better algorithm in the system to increase its performance. There are still many limitations and improvements are to be discover. Until then, hopefully the limitations of the system are to be solved as much as possible, and go through multiple high quality software testing to ensure its reliability and it can be deployed to replace the manual booking process that has been use for about 50 years as early as possible to serve and give more convenience to TAR UC's academic staffs, students and alumni.

6.2 Achievement

There are a few objectives that has been accomplished by implementing the online Facility Booking System, they are:

1. Migrating the facility booking process from manually to online.
 - The manual way of facility booking are to be considered as outdated because of the slow and inefficient processing speed using paper as media to record all the booking information. Additionally the manual process unable to catch up with the modern fast-paced society that demanding high performance and efficient system.
2. Increase of working efficiencies of management staffs.
 - Staffs able to perform facility booking or search customer's booking record with just a few clicks in the system whereas lots of time are needed for staffs to search through all the paper records just to find a specific record. The difference in efficiency of performing tasks is significant for online Facility Booking System.
3. Provide a better platform for both management and customer to perform facility booking
 - More and more people prefer to use mobile phone as a replacement of anything such as computer, walk-in shopping and etc. Smartphone allows people to perform

lots of tasks yet eliminating the necessity of them to physically approach to a specific location just to accomplish a tasks. Thus mobile phone has become a daily life needs for people.

6.3 Contributions

The online Facility Booking System will bring benefits to three parties: TAR UC's students, staffs and alumni. The system can be deployed to serve all the undergraduates and academic staffs, allowing them to enjoy the benefits and features provided by the system. The system allow staffs to assist student in account management, announcement posting, facility booking and others via webpage platform. This system also helps in reduce the workload of staffs as students will perform facility booking by using mobile app instead of approaching counter to do so.

The system allow staffs to manage the venue, services, staff and customer record, announcement, and booking just within few touches or clicks and information that is required to shown to students can be sent immediately so that students able to receive the latest update of information. Security is also an important factor that must be emphasized for the system to secure all the information stored from being leak.

6.4 Limitations and Future Improvements

Throughout the whole development process, there are few limitations that can be improved and enhancement that can be implemented in the future found.

Limitations

- Password recovery
 - Currently the system will just generate a new random password and change the password of an account, then send an email containing the password to the account's mailbox. This solution is very unsecure as hackers might access to the mailbox to retrieve the password. Instead of doing so, sending a onetime password (OTP) to the account's mailbox and use the OTP to verify status and reset the password is the best way for password recovery.
- Venue
 - The venue retrieved from the database are then resolved into a list of tree. Problems came when then system need to access the child of the tree, recursive loop is

needed for the current solution and this bring a lot of burden for the computer to process and it consumes a lot of resources and effort to design the algorithm. The best solution for the limitations is apply progressive way of retrieve venue, retrieving a list of child venue by passing in the parent venue, then display the child venue out.

- Add Customer
 - Currently the only way for staff to add customer is by importing a list of customer record stored in json file. Staffs has no authority to assist customer to activate customer's account. Instead of just importing json file, customer should also able to create customer account one by one with inactive status, then let customer to verify their account. Or allow staffs to assist customer in activating their account.

6.5 Issues and Solutions

Various kinds of problem was faced throughout the whole project development period. First and foremost is the lack of documentation support of the system. Although the developers that working on the system do have some documentation support, but the information of the documentation is very simple, and missing lots of important information such as database expression. Lacks of documentation support leads to extra time spent in analyzing the system. The solutions is spending more time to further analyze the system and produce a complete documentation for other developers to make their analyzing process easier.

Besides, the final version of system left by previous developers are full of flaws. Most of the modules are not working correctly. Extra effort is put inside to identify the problems and fixing it.

Furthermore, the framework used for the system is a mix and match of Spring MVC. The system do use the feature from Spring MVC, but did not completely implementing it. This makes the process of importing and syncing libraries become very difficult. This can be solved by completely implementing Spring MVC into the system.

Last but not least, the server that system is deploy in is facing some issues and came offline. The developer that working on mobile app version of Facility Booking System could not communicate with the server and retrieve data. A new temporary server is set-up for the developer to retrieve data and complete the development. Luckily the server came online at last.

References

- Akoum, I. (2016) 'RESEARCH , DEVELOPMENT AND INNOVATION IN MALAYSIA : ELEMENTS OF AN EFFECTIVE GROWTH MODEL Contribution / Originality', 6(7), pp. 390–403. doi: 10.18488/journal.aefr/2016.6.7/102.7.390.403.
- Al-maktari, O. A. M., Jamaludin, R. and Hosam, A. (2012) 'The Acceptance of Online Booking System (OBS) Based on the Theory of Reasoned Action (TRA): A Case of Sana ' a University', 3(2), pp. 3–6.
- Byrne, M. (2001) 'Interviewing as a data collection method.', AORN journal, 74(2), pp. 233–235. doi: 10.1016/S0001-2092(06)61533-0.
- Chen, Y. C. et al. (2017) 'Cloud-fog computing for information-centric Internet-of-Things applications', Proceedings of the 2018 IEEE International Conference on Applied System Innovation: Applied System Innovation for Modern Technology, ICASI 2017, pp. 637-640. doi: 10.1109/ICASI2017.7988506.
- Fletcher, Anthony. (1986). Computerizing records from Leicestershire's museums. Meseum Documentation System, 182-189.
- Liao, K. and Lee, W. (2010) 'A Novel User Authentication Scheme Based on', 5(8), pp. 937–941. doi: 10.4304/jnw.5.8.937-941.
- MacRoberts, M. H. and MacRoberts, B. R. (2010) 'Problems of citation analysis: A study of uncited and seldom- cited influences', Journal of the American Society for Information Science and Technology. Wiley Online Library, 61(1), pp. 1–12.
- Molchanova, V. S. (2014) 'European Journal of Social and Human Sciences , 2014, Vol.(2), № 2', pp. 75–80.
- Osman, M. A. et al. (2012) 'A Study of the Trend of Smartphone and its Usage Behavior in Malaysia', International Journal on New Computer Architectures and Their Applications (IJNCAA), 2(1), pp. 275–286. doi: 10.1007/978-3-642-22603-8_35.
- Proceedings, S. (2011) 'QR Codes : What Are They and Why Should You Care ?', 1(1), pp. 0–8.
- Twenty-one, S. B. N. (2017) 'INTERNET USERS SURVEY 2017'.

Walsh, A. (2009) 'Quick response codes and librarios', Library Hi Tech News, 26(5/6), pp 7-9. doi: 10.1108/074190509109852551.

What Is Software Testing – Definition, Types, Methods, Approaches. (17 March, 2019). Retrieved from Software Software Testing Material: <https://www.softwaretestingmaterial.com/software-testing/>

Appendix`

Proposal

Abstract

In such a modern era where technology is getting more and more advanced, there is a trend where people starts to looking for and more emphasized on an efficient and effective method that able to bring a lot of convenience to people. The efficient and effective method that could bring more convenience to people includes software that could be found such as Joan's Meeting Board, a Facility Booking System (FBS), and so much more. Following the introduction of the Internet in November 1995, Internet users have increased significantly to date, occupying 76.9 percent of Malaysia 's population, making Malaysia an Internet-oriented society(Twenty-one, 2017), most of which have smartphone access to the Internet, which is the current trend in Malaysia (Osman *et al.*, 2012). Unfortunately, it is hard to say that everyone is able to adapt to the rapidly changes of technology trend and applying those advance technology to everywhere. There are still some places which unable to catch up to follow the trend and retain the traditional method to perform tasks. But luckily Tunku Abdul Rahman University College (TARUC) is on the way catching up with the technology trend. College students are putting their effort creating a facility booking system to make the facility booking process and experience more convenient. The existing booking system done allows admin user to make facility booking, customer and staff account managemnt, event management, and make annoucement. However, there are some vulnarabilities left behind by students that done this project and also still much room for further enhancement andtowards the existing facility booking system done. The proposed enhancement allows admin user to plan the schedule of facility and resouces or venue availability before allows customer to make booking, control the accesibility of admin user with predefined role, maintaining customers' record, additional api support for mobile development and vulnarabilities fixing for smooth facility booking, announcement management, and service and rules management experience.

Problem

In today's world, efficient and effective method that able to help in produce high productivity with much lower cost compared to traditional ways started to emphasize by

peoples due to the increasing in speed towards pace of life. This trend is obvious to be seen especially in Malaysia, which is a developing third-world country. The country is gradually shifting away from agrarian economy to a more research and development (R&D) and technology related activities (Akoum, 2016). Despite more and more advanced technology was introduced in Malaysia, there are still lots of places in Malaysia which yet to adapt or assimilate into the technology-advance environment and still using traditional method to work and so on. But luckily in Tunku Abdul Rahman University College (TARUC), the college has started to adapt to the technology trend and implemented different kinds of technology such as Facility Booking System (FBS). The facility booking system currently allows admin and user to make facility booking using their smart phone via internet, announcement and event management, and customer and staff account management. However, the functionality provided was very basic, further development and enhancement are to be expected to increase the user experience. The existing system lacks of management towards the scheduling of facilities and resources before allowing customer to make booking. Customers are required to login into the system, but the user account having limited amount of personal information. Complete personal information allows admin to contact customer easily. Besides, additional security should be apply towards admin account while they are using the system as the role of each admin are different. Furthermore, additional support towards mobile platform from the system is needed to provide smoother booking experience to the customers. Last but not least, there are over 90 percent of the system experiencing vulnerabilities when the project is handed over. Long story short, these are the few problem that was identified and expected to be solved to provide better user experience.

Solution

To resolve problems identified as stated above, a list of enhancement has been proposed. The proposed enhancement able to further improve the efficiency of facility scheduling, effectively control the accessibility of admin user. The proposed enhancement allows admin to have full control towards all the facility schedule with independent availability. Admin able to create, edit and delete the facility schedule and with some control such as maximum duration of each booking. Admin also able to have a better view on all the facility with the UI support. Besides, different admin user has its own role and responsibility towards the usage of the system. Thus role restriction are added into the admin module to prevent admin

from accessing modules that he/she are not allowed to access. In addition, admin will have better customer's information management and support when they need to do so to provide service to customer. Additional effort was invested in fixing the vulnerabilities of facility booking, announcement management and service and rules management. To sum up, this proposed system can certainly solve the problems currently faced by staff, replacing the current manual booking system and making improvements that have already proved useful, easy to use and satisfactory (Al-maktari, Jamaludin and Hosam, 2012).

Target Market

The application is mainly targeting users which is in tertiary educational institution. User such as students, staffs and alumni will be the most beneficial target from the proposed system because the necessity of approaching respective facility booking counter was eliminated. It was quite frustrated when approaching counters and found that the booking available slot left does not fits own schedule. Furthermore, users are lazy to go to booking counter make their booking. Additionally, student need to walk far distance to the booking counter. User to access the booking system by using the mobile application (front-end) to make their facility booking. The booking can be done anywhere within the predefined available time-frame. User also able to approach booking counter to make their booking where admin will help user to perform booking process via web application (back-end).

Competition/Contribution

In traditional facility booking process, students and staffs are required to physically approach the booking counter to make any facility booking with related staffs at the counter. With the proposed system and enhancement applied, the time spent on facility booking are expected to reduce in a significant figure. Certainly there are always competitors in the society, company such as Planyo are focusing on automated system for their booking system that require zero interaction with human-beings and yet the physical access of the system do not require any internet connection to access certain place customer booked remotely. Besides, there are also company that was succeed in the facility booking areas such as Joan Meeting Boards. The setup process for the boards is simple, yet the scheduling is directly linked to google calendar. Any booking made on google calendar will be updated immediately.

Milestones

Table A1: Milestone

Phase	Milestone	Deadline
System Planning	First draft proposal submission	05/06/2018
	Final proposal submission	22/06/2018
System Analysis	Introduction	13/07/2018
	Research background	10/08/2018
	Methodology & Requirement analysis	15/10/2018
System Design	Front-End System App UI Design (Students)	17/10/2018
	Front-End System Database design	25/10/2018
	Front-End System Module design (Students)	31/10/2018
	Submission of Project I Portfolio	16/11/2018
System Development	Front-End System App Coding (Students)	18/11/2018
	Unit Testing (Students)	15/01/2019
	Module Testing (Students)	17/01/2019
	Integration Testing	09/02/2019
	Final System Testing	04/03/2019
	Submission of Draft FYP Report	29/03/2019
	Submission of Final FYP Report	05/04/2019

Appendix Proposal and Chapter 1

Module Description

- Backend(Admin) Module

This module is capable to handle the general court booking workflow such as check schedule available, add friends, check announcement and etc.

- Backend(Admin) Module

This module allow facility manager/staff to monitor and control the availability of facility opened for booking, control user's accessibility and create events.

Table A2: Module Description

Module Name	Platform	Description
Admin	Web	• Restrict admin to access module based on his/her role assigned. • Allow admin to reset their password
Announcement	Web	• Vulnerability fixing
Booking	Web	• Vulnerability fixing. • Algorithm changing for additional filtering
Customer	Web	• Allow admin to help customer perform account registration. • User profile to provide more user information. • To allow admin to import a list of customer record. • Allow customer or admin to perform password reset
Rules and Service	Web	• Vulnerability fixing
Venue	Web	• Used to store facility and resources information before allowing the scheduling of facility. • Allow staff to manage the operating hour of the venue

Table A3: Workload Distribution

Module and Sub-modules	Melvin Low Eik Kit	Ong Zhe Zhun
Front End	✓	
Authentication	✓	
Announcement	✓	
Schedule booking	✓	
Booking Record	✓	
Friend system	✓	
Back End		✓
Admin		✓
Announcement		✓
Booking		✓
Customer		✓
Rule and Service		✓
Venue		✓

Table A4: Tool Used

Tool name	Description
sql database	A database that stores all the information such as user information, facility booking details and schedule, event details and etc. Sql database are used because it was setup inside trusted machine and network connection was secured by taruc's network administrator.
Server-sized computer	The server computer is used to host the sql database, web application and also the hosting of mobile application server so that server request from mobile user able to communicate with database server.

Netbeans IDE	Netbeans IDE is the tool used to perform editing of the back-end code. It provides good support towards compiling, debugging of codes.
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Appendix Chapter 2

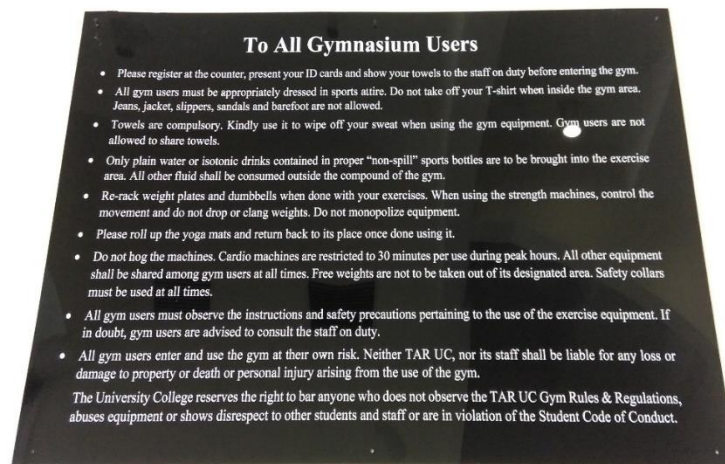


Figure A1: Rules and Regulations in Gymnasium

TUNKU ABDUL RAHMAN UNIVERSITY COLLEGE
COURTS BOOKING
MONDAY
DATE: 29/11/17

TIME	9:00-10:00	10:00-11:00	11:00-12:00	12:00-1:00	1:00-2:00	2:00-3:00	3:00-4:00	4:00-5:00	5:00-6:00	6:00-7:00	7:00-8:00	8:00-9:00	AMOUNT (RM)
Gymnastics	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Badminton	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Tennis	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Basketball	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Table Tennis	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Volleyball	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Squash	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Futsal	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	
Snooker	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	CO-ED	

Figure A2: Traditional Booking Paper Used by Staffs



Figure A3: PC That Was Used by Staffs to Enter Booking Record

SPORTS FACILITIES RATES IN TAR COLLEGE FOR FAMILY ME

NO	FACILITY	TIME			S
		MONDAY - FRIDAY	SATURDAY	SUNDAY	
1	Basketball Court	9am - 9pm	9am - 7pm		
2	Volleyball Court	9am - 9pm	9am - 7pm		
3	Netball Court	9am - 9pm	9am - 7pm		
4	Tennis Court	9am - 9pm	9am - 7pm		
5	Table Tennis	9am - 9pm	9am - 7pm		
6	Squash Court	9am - 9pm	9am - 12pm 3pm - 7pm	3pm - 7pm	
7	Badminton Court	9am - 9pm	9am - 7pm		
8	Football Field	9am - 7pm	9am - 7pm		
9	Futsal Court	9am - 7pm	9am - 7pm		
10	Gymnasium 1 (Club House)	9am - 11am 11am - 1pm 1pm - 3pm 3pm - 5pm 5pm - 7pm 7pm - 9pm	9am - 11am 3pm - 5pm 5pm - 7pm	3pm - 5pm 5pm - 7pm	
11	Gymnasium 3 (Sports Complex)	9am - 11am 11am - 1pm 1pm - 3pm 3pm - 5pm 5pm - 7pm 7pm - 9pm	9am - 11am 11am - 1pm 1pm - 3pm 3pm - 5pm 5pm - 7pm		
	Swimming Pool	9am - 12pm 12pm - 5pm	9am - 12pm 3pm - 7pm	3pm - 7pm	

Figure A4: Schedule of facility in TARUC

RULES & REGULATIONS FOR USE OF SPORTS FACILITIES:

1. Players should be properly attired at all times. **Shorts and T-shirts must be worn at all times and sports shoes are mandatory.** For **squash and badminton**, wear only **non-marking sports shoes** within the court.
2. All applications to book sports facilities must be submitted at least **one day in advance** except for use of swimming pool and gymnasium. Players should present **booking confirmation slip** together with all the **group participants' ID cards** to the staff on duty before using the facility.
3. For gym equipment use, **towels are compulsory**. Please **register at the counter** and present **your ID card** to the staff on duty before entering the gym. Do not drip sweat on the equipment. Wipe your sweat off the benches after you have used them. Re-rack weight plates and dumbbells when done with your exercises. Do not monopolize equipment. Do not drop or clang weights.
4. **Food and bags are not allowed** at the sports venue. Drinks and water may be taken in non-breakable, spill-proof containers.
5. Players are advised to bring their own equipment. However, players who wish to borrow equipment can refer to the staff on duty. **Any person who damages or destroys the property of the University College shall be held responsible and liable for the repair and replacement of such property at his or her own expense.**
6. The University College reserves the **right to cancel bookings without prior notice.**
7. The University College reserves **the right to bar anyone who does not observe the Rules and Regulations for Sports Facilities**, abuses equipment or shows disrespect to other students and staff or are in violation of the Student Code of Conduct.

Rates and Rules & Regulations are subject to change by University College from time to time. Users may be notified of such changes in any manner deemed appropriate by the University College.

Figure A5: Rules and Regulations for Use of Sports Facilities

Server Guide**Server Specification**

VM Name	Sport Facility
IP Address	vNIC0: 172.18.2.13 vNIC1: 172.18.17.2 (Main) Port 8884 Public IP: 103.52.192.245
VLAN	20 170
GW FG200	NIL 172.18.17.1
Uplink	vmnic0 vmnic3
OS	Windows Server
CPU	4 CPU 4 Core
RAM	8GB
HD	500GB

Remote Desktop Connection

Username	student
Password:	Student@i2

Instructions

1. Search from PC for *REMOTE DESKTOP CONNECTION* application.
2. Open Remote Desktop Connection. On the Computer field, put in the IP address for the VM you would like to access.
3. Click on Show Options to input username to be used
4. Click Connect and it will prompt you for password. Once the connection is successful, you will see a full screen of desktop as if you are in front of the server.

Tomcat

User	taruc
Password	taruc123

MySQL

Root account

Password: taruc53300

TCP/IP

Port: 3306

X Protocol Port: 33060

MySQL User

Username : taruc_dba

Host : localhost

Role : DB Admin

Password : taruc53300

Windows Service Name : MySQL80

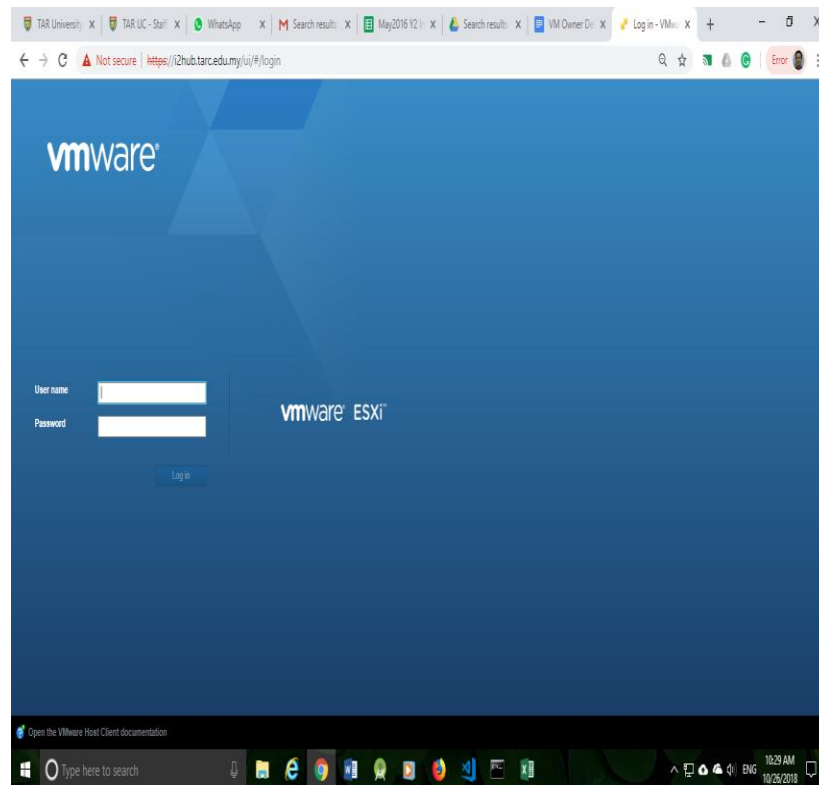
Firewall Setting

To add a rule in the Windows Firewall for port 8884, do the following:

- Go to the Control Panel and launch "Windows Firewall"
- Go to "Advanced Settings"
- Select "Inbound Rules" in the left pane
- Select "New Rule" in the right pane
- In the New Inbound Rule Wizard, select "Port" as Rule Type, then click on "Next"
- Select "TCP and put "8884" (and any other ports you want to open) in "Specific local ports", then click on "Next"
- Select "Allow the connection", then click on "Next"
- Select the network location where the rule should apply (select them all if you're not sure), then click on "Next"
- Give a name and an optional description

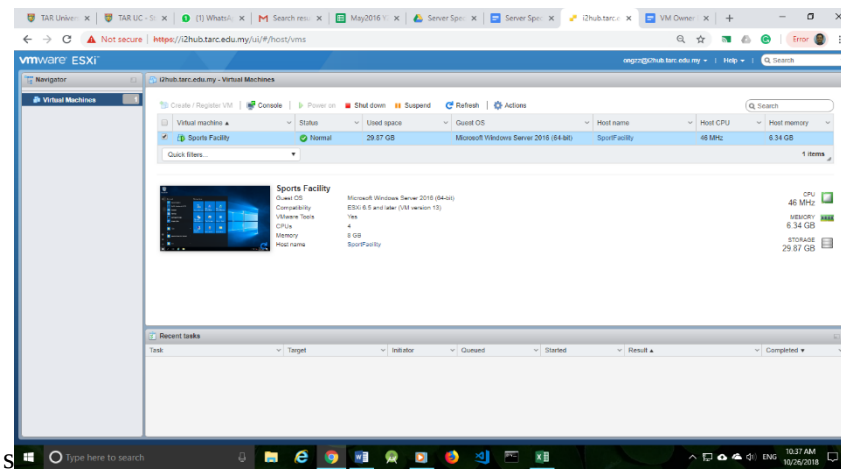
Access VM

1. Using any browser, enter the following URL:
<https://i2hub.tarc.edu.my>
2. When prompt as shown below, enter user name and password.



Username	Password
choylfongzz	VMware1!

3. In the VMware control panel, select the Sports Facility VM and click the screen thumbnail to view the server.

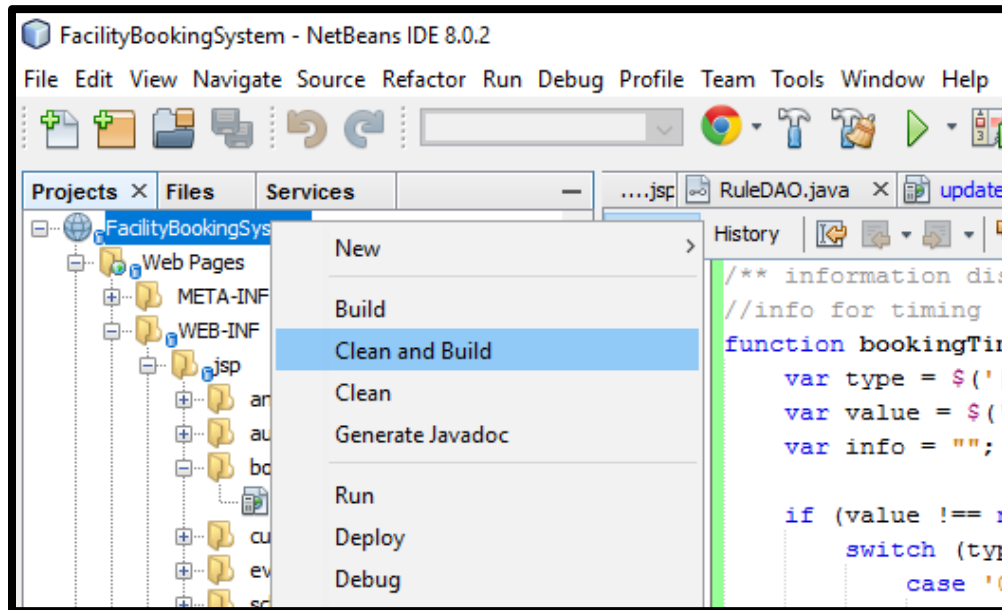


4. To login to the Windows server, select “Administrator” and enter the password “VMware1!”

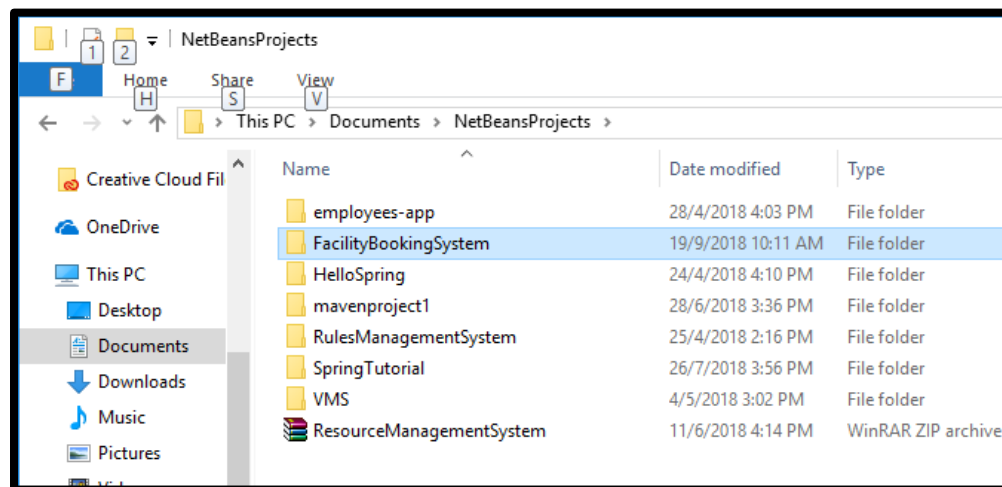
System Deployment Process via WAR File

Step 1: Create war file via NetBeans

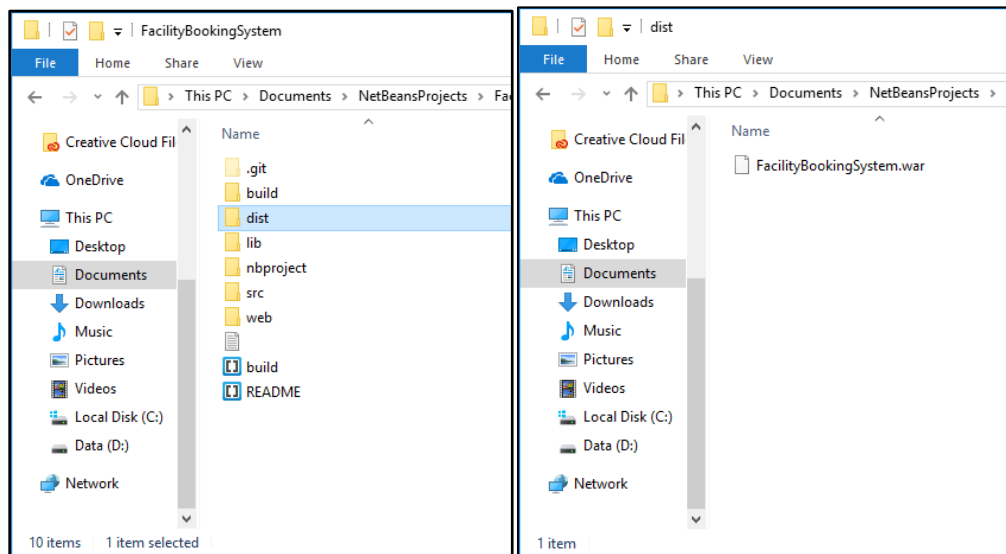
1. Open the project you wanted to deploy in NetBean
2. Right Click on the project name -> Click “Clean and Build”



3. After the process is done, go to your project folder using File Explorer.



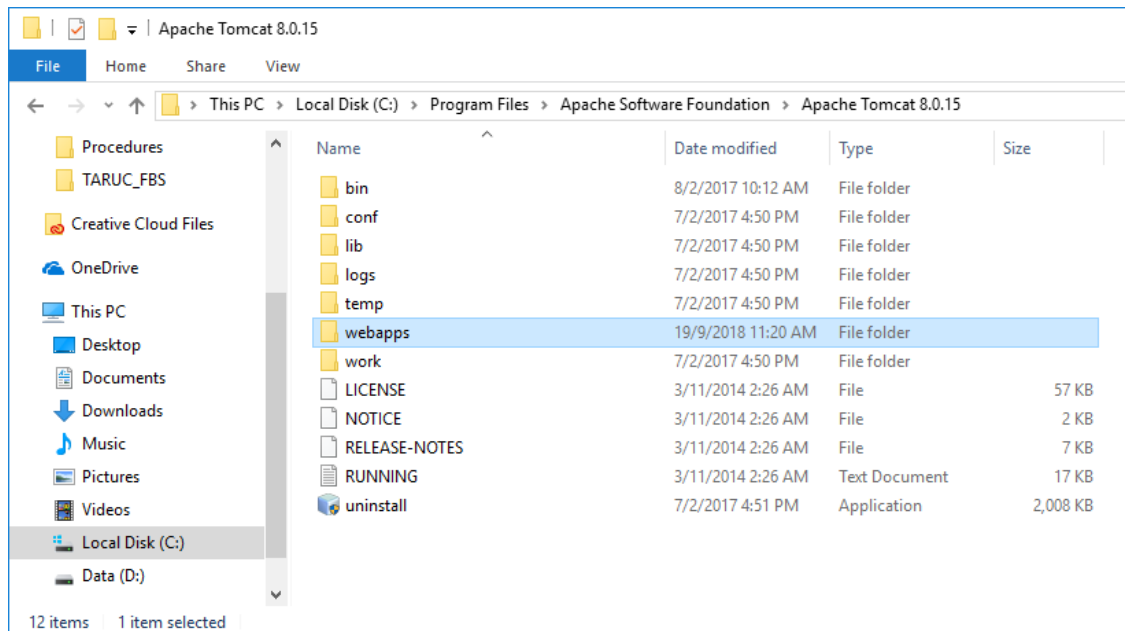
- Then, go into your project file -> dist.



- The war file inside the dist folder will be used to deploy.

Step 2: Deploy war file to the server

- Transfer the war file to the server example FTP, using USB, upload, etc.
- Copy the war file to the tomcat webapps folder.



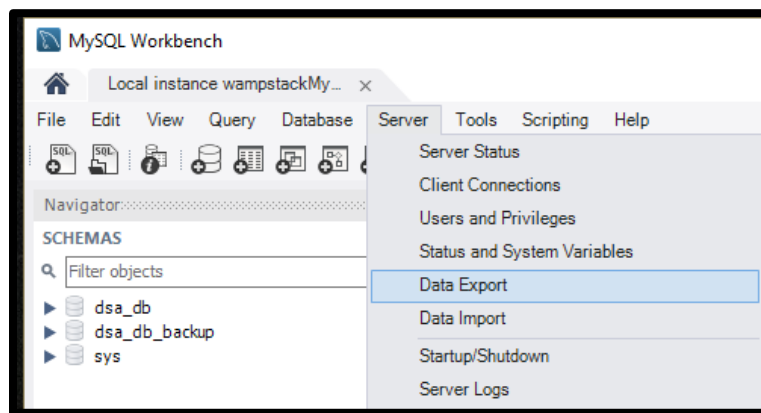
- The war file will be deploying automatically by default.

4. Type in the “localhost:<port>/<project name>” to test if the project is deployed successful.

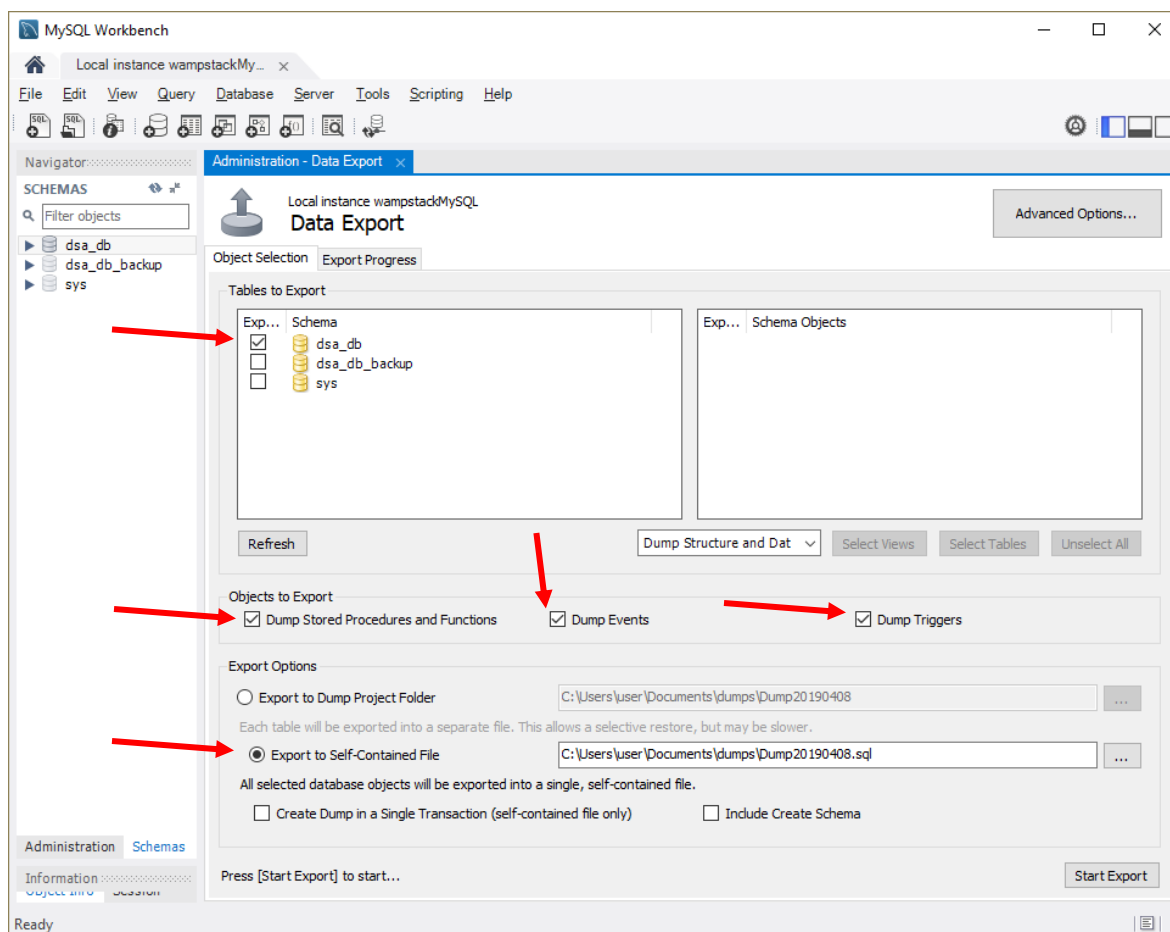
Great reference: <https://medium.com/@hariomvashisth/deploying-java-web-applications-aws-7f8745945646>

Export Database

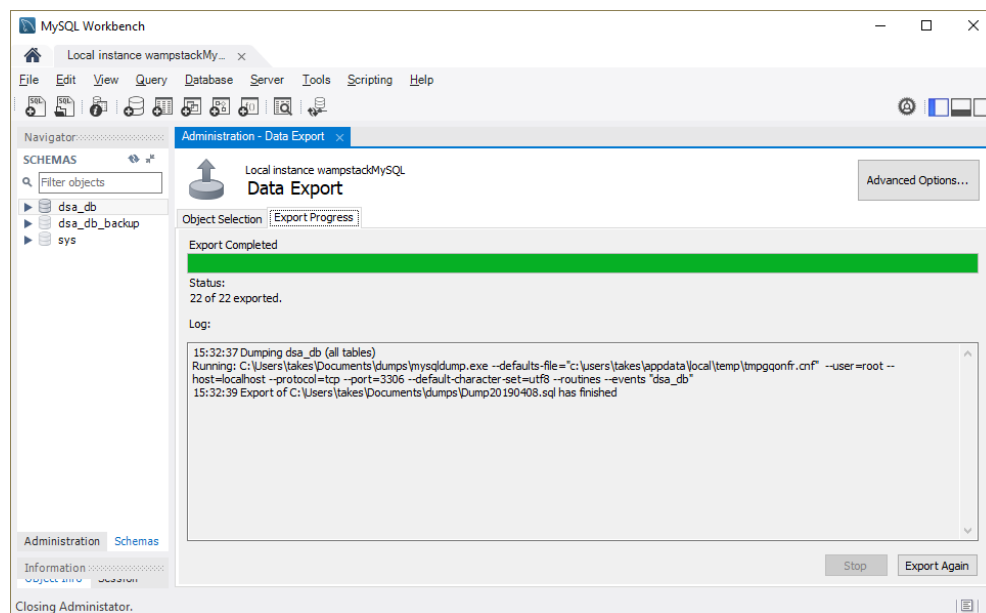
1. Open MySQL Workbench.
2. Login to the database.
3. Click on the “Server” on the menu and select “Data Export”



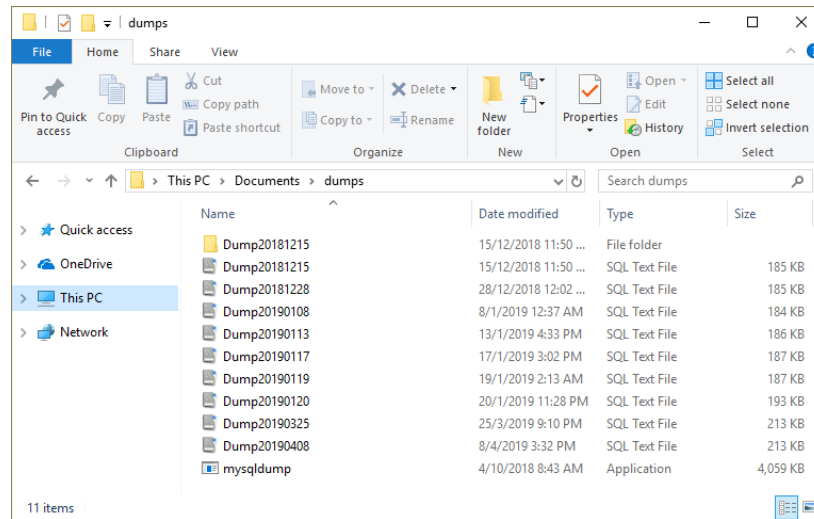
4. In the data export page, make sure to check the tables to export of “dsa_db”, check all the boxes for object to export, and export options of export to self-contained file.



5. Click on “Start Export” on the bottom right to proceed export.
6. Once export completed, the export progress will notify.

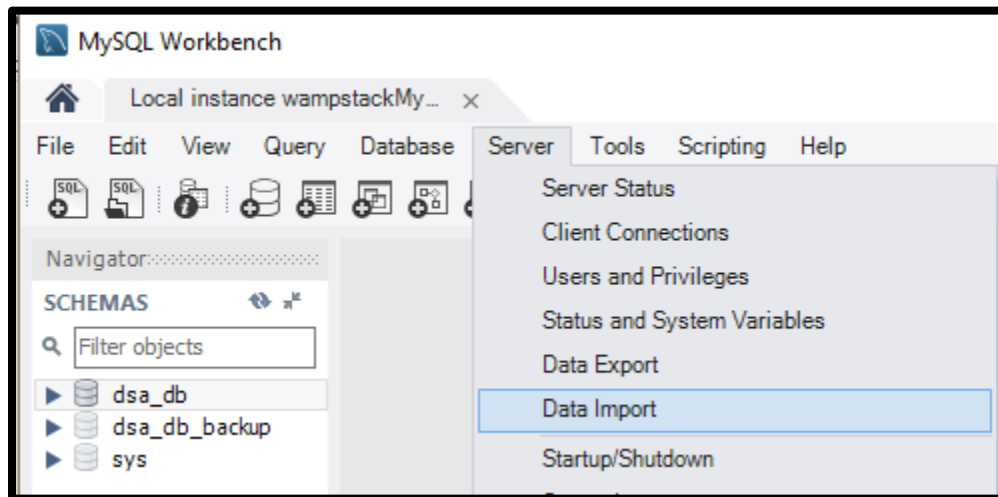


7. The exported file can be found in “C:\Users\<user>\Documents\dumps\”

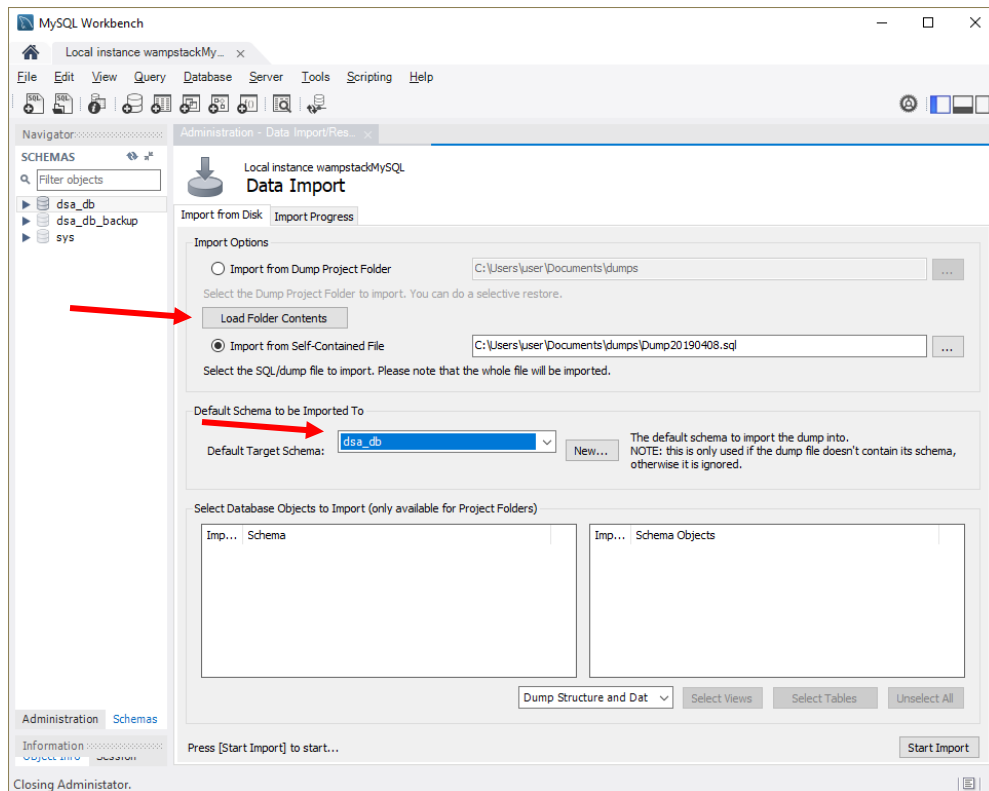


Import Database

1. Open MySQL Workbench.
2. Login to the database.
3. Click on the “Server” on the menu and select “Data Import”



4. In the data import page, make sure to select import options of import from self-contained file, select the desire exported file, select default target schema of “dsa_db”.



5. Click on “Start Import” on the bottom right to proceed import.
6. Once import completed, the import progress will notify and end the import process.

